

Chain of Preference Optimization: Improving Chain-of-Thought Reasoning in LLMs

偏好优化链: 提升大语言模型中的思维链推理

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Abstract

摘要

The recent development of chain-of-thought (CoT) decoding has enabled large language models (LLMs) to generate explicit logical reasoning paths for complex problem-solving. However, research indicates that these paths are not always deliberate and optimal. The tree-of-thought (ToT) method employs tree-searching to extensively explore the reasoning space and find better reasoning paths that CoT decoding might overlook. This deliberation, however, comes at the cost of significantly increased inference complexity. In this work, we demonstrate that fine-tuning LLMs leveraging the search tree constructed by ToT allows CoT to achieve similar or better performance, thereby avoiding the substantial inference burden. This is achieved through Chain of Preference Optimization (CPO), where LLMs are fine-tuned to align each step of the CoT reasoning paths with those of ToT using the inherent preference information in the tree-search process. Extensive experimental results show that CPO significantly improves LLM performance in solving a variety of complex problems, including question answering, fact verification, and arithmetic reasoning, demonstrating its effectiveness. Our code is available at <https://github.com/sail-sg/CPO>.

思维链 (CoT) 解码的最新进展使大语言模型 (LLMs) 能够为复杂问题解决生成明确的逻辑推理路径。然而, 研究表明这些路径并不总是经过深思熟虑且最优的。思维树 (ToT) 方法采用树搜索来广泛探索推理空间, 并找到 CoT 解码可能忽略的更好的推理路径。然而, 这种深思熟虑是以显著增加推理复杂性为代价的。在这项工作中, 我们证明利用 ToT 构建的搜索树对 LLMs 进行微调可以使 CoT 实现相似或更好的性能, 从而避免巨大的推理负担。这是通过偏好优化链 (CPO) 实现的, 在 CPO 中, 利用树搜索过程中的固有偏好信息对 LLMs 进行微调, 使 CoT 推理路径的每一步与 ToT 的推理路径对齐。大量实验结果表明, CPO 在解决包括问答、事实验证和算术推理在内的各种复杂问题时显著提高了 LLMs 的性能, 并证明了其有效性。我们的代码可在 <https://github.com/sail-sg/CPO> 获取。

1 Introduction

1 引言

Recent advances in large language models (LLMs) have shown that constructing reasoning chains is critical to improving their problem-solving capabilities [1, 2, 3, 4, 5, 6, 7]. A representative method is chain-of-thought (CoT) [1], which prompts LLMs to generate intermediate reasoning steps, i.e., thoughts, thereby constructing explicit reasoning paths (as depicted in Figure 1(a)). While straightforward and intuitive, recent research observes that CoT can often overlook optimal reasoning paths and exhibit an unconscious style of answering due to its single-path focus [8,9]. To foster a more deliberate and conscious reasoning style, Yao et al. [8] propose tree-of-thought (ToT), which generates multiple branching thoughts at each step of the reasoning process and conducts self-evaluation for pruning and planning to search for reasoning paths (as shown in Figure 1(b)). However, despite improving reasoning quality, ToT significantly increases computational complexity, which limits its practical application. This raises the question: Can the strategic depth of ToT be integrated into CoT to enhance its effectiveness while maintaining efficiency?

大语言模型 (LLMs) 的最新进展表明, 构建推理链对于提高其解决问题的能力至关重要 [1, 2, 3, 4, 5, 6, 7]。一种具有代表性的方法是思维链 (CoT)[1], 它促使大语言模型生成中间推理步骤, 即思维, 从而构建明确的推理路径 (如图 1(a) 所示)。虽然简单直观, 但最近的研究发现, 由于 CoT 专注于单一路径, 它常常会忽略最优推理路径, 并表现出无意识的回答方式 [8,9]。为了培养更审慎和有意识的推理风格, Yao 等人 [8] 提出了思维树 (ToT), 它在推理过程的每个步骤生成多个分支思维, 并进行自我评估以进行修剪和规划, 从而搜索推理路径 (如图 1(b) 所示)。然而, 尽管提高了推理质量, 但 ToT 显著增加了计算复杂度, 这限制了其实际应用。这就提出了一个问题: 能否将 ToT 的策略深度融入 CoT, 以在保持效率的同时提高其有效性?

Existing research has initially provided a positive answer to the above question [10, 11, 12]. A natural strategy is to treat the reasoning path discovered by ToT for each instance as a target for supervision, and then fine-tune LLMs to improve their CoT reasoning abilities [11, 12]. Several methods have been proposed to improve this approach, including using advanced tree-search techniques like Monte

现有研究最初对上述问题 [10, 11, 12] 给出了肯定答案。一种自然的策略是将思维树 (ToT) 为每个实例发现的推理路径作为监督目标, 然后微调语言模型以提高其思维链 (CoT) 推理能力 [11, 12]。已经提出了几种方法来改进这种方法, 包括使用像蒙特这样的先进树搜索技术

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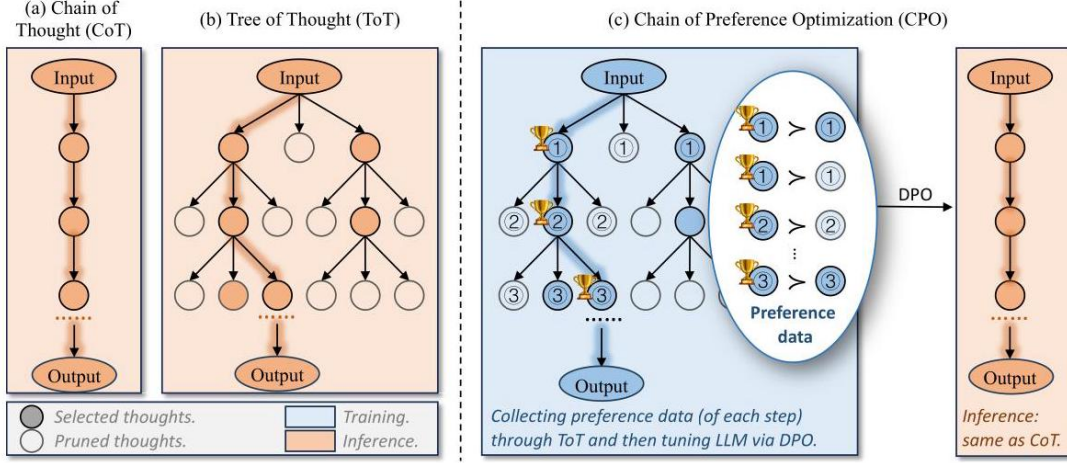


Figure 1: Comparison of CoT, ToT, and CPO methods, where each node illustrates a step in the reasoning process, forming coherent language sequences aimed at solving a problem. The highlighted path indicates the chosen reasoning trajectory. In the CoT method, the LLM generates only one new node at each step, and all generated nodes are used to build the final reasoning path. For ToT, the LLM produces k new nodes at each step, but only the top n -best nodes are kept, with the rest being pruned. In CPO, nodes marked with a trophy represent preferred thoughts, while those marked with numbers are nodes that can be utilized to create preference data. This method uses the search tree structure from ToT to develop paired preference data, subsequently training LLMs to align with these preferences through DPO.

图 1: 思维链 (CoT)、思维树 (ToT) 和偏好优化 (CPO) 方法的比较, 其中每个节点表示推理过程中的一个步骤, 形成旨在解决问题的连贯语言序列。突出显示的路径表示所选的推理轨迹。在思维链方法中, 语言模型在每个步骤仅生成一个新节点, 并且所有生成的节点都用于构建最终的推理路径。对于思维树, 语言模型在每个步骤生成 k 个新节点, 但仅保留前 n 个最佳节点, 其余节点被修剪。在偏好优化中, 标有奖杯的节点表示首选思维, 而标有数字的节点是可用于创建偏好数据的节点。此方法使用思维树的搜索树结构来开发成对的偏好数据, 随后通过基于偏好的优化 (DPO) 训练语言模型以与这些偏好对齐。

Carlo tree-search (MCTS) and employing external reward models [12, 10] for pruning and planning to gather better reasoning paths as supervision. The effectiveness of these approaches is therefore largely dependent on the quality of the best-discovered reasoning path.

蒙特卡洛树搜索 (MCTS) 并采用外部奖励模型 [12, 10] 进行剪枝和规划, 以收集更好的推理路径作为监督。因此, 这些方法的有效性在很大程度上取决于所发现的最佳推理路径的质量。

In this paper, we identify a limitation in these approaches: they overlook the non-optimal reasoning thoughts generated during the tree-search process, which naturally provides additional preference information. Specifically, ToT inherently generates multiple alternative thoughts at each reasoning step, and pruning is performed according to their evaluated qualities. This tree-search process constitutes a preference over all intermediate thought candidates—thoughts appearing in the best-discovered reasoning path are preferred over those that do not. Moreover, this could shed even more insights than the final best-discovered reasoning path, as non-optimal reasoning paths (and thus preferences) exist at each step in the tree-search.

在本文中，我们识别出这些方法中的一个局限性：它们忽略了在树搜索过程中产生的非最优推理思路，而这些思路自然地提供了额外的偏好信息。具体而言，思维树 (ToT) 在每个推理步骤中都会固有地生成多个替代思路，并根据它们的评估质量进行剪枝。这个树搜索过程构成了对所有中间思维候选的一种偏好——出现在最佳发现的推理路径中的思路比那些不在该路径中的思路更受青睐。此外，这可能比最终最佳发现的推理路径提供更多的见解，因为在树搜索的每个步骤中都存在非最优推理路径（以及相应的偏好）。

Inspired by recently developed reinforcement learning from human feedback (RLHF) techniques like direct preference optimization (DPO) [13], we propose Chain-of-Preference Optimization (CPO) to fully exploit the inherent preference information. Specifically, we construct paired preference thoughts at each reasoning step according to the search tree of ToT and then train LLMs to align with these preferences using the DPO algorithm (as illustrated in Figure 1(c)). The paired preference thoughts are constructed based on the above intuition: at each reasoning step, we categorize thoughts as preferred or dispreferred based on their inclusion in the final paths chosen by ToT. With such preference data, CPO enables LLMs to generate the path preferred by ToT using CoT decoding at inference time.

受近期开发的人类反馈强化学习 (RLHF) 技术 (如直接偏好优化 (DPO)[13]) 的启发，我们提出了偏好链优化 (CPO)，以充分利用固有的偏好信息。具体来说，我们根据思维树 (ToT) 的搜索树在每个推理步骤构建成对的偏好思路，然后使用 DPO 算法训练大语言模型 (LLMs) 以与这些偏好对齐 (如图 1(c) 所示)。成对的偏好思路是基于上述直觉构建的：在每个推理步骤中，我们根据思路是否包含在思维树 (ToT) 选择的最终路径中，将其分类为偏好或非偏好。有了这样的偏好数据，CPO 使大语言模型 (LLMs) 能够在推理时使用思维链 (CoT) 解码生成思维树 (ToT) 偏好的路径。

We conduct extensive experiments to evaluate the effectiveness of CPO. Experiments on seven datasets using LLaMA [14] and Mistral [15] as base models demonstrate that CPO is highly effective in teaching LLMs the preferred thoughts of ToT at each reasoning step, leading to an average accuracy improvement of up to 4.3% compared to the base models. Additionally, the experiments reveal that CPO can achieve comparable or even superior performance to the ToT method, which on average requires more than 50 times longer for inference.

我们进行了广泛的实验来评估 CPO 的有效性。使用 Llama [14] 和 Mistral [15] 作为基础模型在七个数据集上进行的实验表明，CPO 在教导大语言模型 (LLMs) 在每个推理步骤中思维树 (ToT) 的偏好思路方面非常有效，与基础模型相比，平均准确率提高了高达 4.3%。此外，实验还表明，CPO 可以实现与思维树 (ToT) 方法相当甚至更优的性能，而思维树 (ToT) 方法平均推理时间要长 50 多倍。

2 Related Work

2 相关工作

Reasoning with LLMs. LLMs have been shown to perform better when prompted to engage in multi-step reasoning [1, 2, 3]. Many studies have focused on improving the generated reasoning paths by post-editing [16] or accessing external knowledge [3, 17]. A distinct approach, more relevant to our interests, transforms the linear reasoning structure into a non-linear format such as a tree or graph [18, 19, 20, 8, 9, 21], which combines thought evaluation with search algorithms like depth-first search (DFS) [22]. Different from our

proposed CPO, these methods require searching during inference, which significantly increases latency.

使用大语言模型 (LLMs) 进行推理。当被提示进行多步推理时, 大语言模型 (LLMs) 已被证明表现更好 [1, 2, 3]。许多研究都集中在通过后编辑 [16] 或访问外部知识 [3, 17] 来改进生成的推理路径。一种与我们的兴趣更相关的独特方法是将线性推理结构转换为非线性格式, 如树或图 [18, 19, 20, 8, 9, 21], 它将思维评估与深度优先搜索 (DFS)[22] 等搜索算法相结合。与我们提出的 CPO 不同, 这些方法在推理过程中需要搜索, 这显著增加了延迟。

LLM self-improving. Reinforcement learning (RL) has increasingly been applied to LLMs by treating them as RL agents for alignment with human feedback [23, 24, 25, 26]. Recent advances demonstrate the potential of using LLMs for self-generating data to augment fine-tuning processes [27, 28, 29, 30, 31, 32]. For instance, reinforced self-training methods [33, 34, 35, 36, 37] introduce mechanisms to curate new high-quality examples and iteratively enrich the training dataset for enhancing model performance. Nevertheless, these methods typically rely on either an external reward model [35, 34] or labeled data [33]. In contrast, approaches like self-rewarding [29, 38] utilize LLMs themselves to evaluate the generated content, aligning more closely with our method. However, these strategies still require initial seed data [29, 38], necessitating human annotation. Our work differs from previous methods as it does not rely on any ground-truth data, allowing LLMs to self-learn from their own feedback. Additionally, our approach constructs feedback in a chain fashion, focusing on reasoning steps, an aspect overlooked by prior works.

大语言模型 (LLMs) 自我改进。强化学习 (RL) 越来越多地应用于大语言模型 (LLMs), 将它们视为与人类反馈对齐的强化学习智能体 [23, 24, 25, 26]。最近的进展展示了使用大语言模型 (LLMs) 自我生成数据以增强微调过程的潜力 [27, 28, 29, 30, 31, 32]。例如, 强化自我训练方法 [33, 34, 35, 36, 37] 引入了筛选新的高质量示例并迭代丰富训练数据集以提高模型性能的机制。然而, 这些方法通常依赖于外部奖励模型 [35, 34] 或标记数据 [33]。相比之下, 自我奖励 [29, 38] 等方法利用大语言模型 (LLMs) 本身来评估生成的内容, 与我们的方法更接近。然而, 这些策略仍然需要初始种子数据 [29, 38], 这需要人工标注。我们的工作与以前的方法不同, 因为它不依赖任何真实数据, 允许大语言模型 (LLMs) 从自身反馈中自我学习。此外, 我们的方法以链的方式构建反馈, 专注于推理步骤, 这是先前工作中被忽视的一个方面。

Monte Carlo tree-search for LLMs. Monte Carlo tree-search (MCTS) is a robust algorithm for navigating complex decision-making environments, commonly employed in strategic board games such as AlphaGo [39, 40, 41, 42, 43]. MCTS methodically constructs a search tree, balancing exploration and exploitation, simulates various outcomes, and updates utility estimates based on these simulations. Recent studies have shown that MCTS can enhance the decoding process in LLMs [11, 44, 45, 21, 12]. However, the primary challenge with MCTS is the high latency during inference, particularly in difficult reasoning tasks [46, 47]. While some approaches have attempted to optimize LLMs by leveraging reasoning paths identified through MCTS [11, 12], these methods still rely on labeled data and require separate policy and value models to explore and evaluate potential moves at the tree's leaves. In contrast, our CPO approach eliminates the need for human annotations and simplifies the tuning of LLMs without the necessity for additional models.

用于大语言模型的蒙特卡洛树搜索。蒙特卡洛树搜索 (MCTS) 是一种用于在复杂决策环境中导航的强大算法，常用于诸如 AlphaGo 等策略棋盘游戏 [39, 40, 41, 42, 43]。MCTS 有条不紊地构建搜索树，平衡探索和利用，模拟各种结果，并根据这些模拟更新效用估计。最近的研究表明，MCTS 可以增强大语言模型中的解码过程 [11, 44, 45, 21, 12]。然而，MCTS 的主要挑战在于推理过程中的高延迟，特别是在困难的推理任务中 [46, 47]。虽然一些方法试图通过利用通过 MCTS 识别的推理路径来优化大语言模型 [11, 12]，但这些方法仍然依赖标记数据，并且需要单独的策略和价值模型来探索和评估树的叶子节点处的潜在移动。相比之下，我们的 CPO 方法无需人工标注，并且无需额外模型即可简化大语言模型的调整。

3 Background

3 背景

In this section, we formalize our notation and provide a brief overview of key prior knowledge for our method. We denote language sequences by lowercase letters, e.g., x, y, z , to represent a sequence of tokens. The output distribution of an LLM parameterized by θ is denoted by π_θ .

在本节中，我们形式化我们的符号，并简要概述我们方法的关键先验知识。我们用小写字母表示语言序列，例如， x, y, z ，以表示一系列令牌。由 θ 参数化的大语言模型的输出分布用 π_θ 表示。

3.1 Chain-of-Thought Prompting

3.1 思维链提示

Chain-of-thought (CoT) [1] is a method that prompts LLMs to generate a chain of reasoning steps before the final answer, as shown in Figure 1. It introduces a series of intermediate thoughts, denoted as $z_1, \dots, z_n$, that link an input x to an output y , where n is the total number of reasoning steps. For instance, if x is a combination of demonstration examples and the input question and y is the final answer, each intermediate thought z_i forms a coherent language sequence representing a part of the overall reasoning path toward the final answer. The demonstration examples consist of a set of CoT demonstrations, which serve as exemplars in the prompting process. The intermediate reasoning thoughts are sequentially sampled from the distribution $z_i \sim \pi_\theta(\cdot \mid x, z_1, \dots, z_{i-1})$ and the output is then derived from $y \sim \pi_\theta(\cdot \mid x, z_1, \dots, z_n)$.

思维链 (CoT)[1] 是一种促使大语言模型在给出最终答案之前生成一系列推理步骤的方法，如图 1 所示。它引入了一系列中间思维，记为 $z_1, \dots, z_n$ ，这些思维将输入 x 与输出 y 联系起来，其中 n 是推理步骤的总数。例如，如果 x 是示范示例和输入问题的组合，而 y 是最终答案，那么每个中间思维 z_i 形成一个连贯的语言序列，代表通向最终答案的整体推理路径的一部分。示范示例由一组思维链示范组成，它们在提示过程中作为示例。中间推理思维从分布 $z_i \sim \pi_\theta(\cdot \mid x, z_1, \dots, z_{i-1})$ 中顺序采样，然后从 $y \sim \pi_\theta(\cdot \mid x, z_1, \dots, z_n)$ 得出输出。

3.2 Tree-of-Thought Prompting

3.2 思维树提示

Tree-of-thought (ToT) [8] enables LLMs to explore multiple reasoning paths before answering a given question, as illustrated in Figure 1. This approach models the LLM reasoning task as a search over a tree, where each node represents a thought step in the reasoning path. ToT comprises two main components, both implemented through prompting LLMs: 1) the thought generator and 2) the state evaluator. The thought generator constructs several new thoughts for the next step based on the current state. Subsequently, the state evaluator generates scores for each new thought and selects the n -best thoughts for further search. The final result is determined by the search algorithm (e.g., BFS or DFS) applied over the selected thoughts until the reasoning process reaches a conclusion.

思维树 (ToT)[8] 使大语言模型能够在回答给定问题之前探索多条推理路径，如图 1 所示。这种方法将大语言模型推理任务建模为在一棵树中的搜索，其中每个节点代表推理路径中的一个思维步骤。思维树包括两个主要组件，均通过提示大语言模型来实现：1) 思维生成器和 2) 状态评估器。思维生成器根据当前状态为下一步构建几个新思维。随后，状态评估器为每个新思维生成分数，并选择 n 个最佳思维进行进一步搜索。最终结果由应用于所选思维的搜索算法（例如，广度优先搜索或深度优先搜索）确定，直到推理过程得出结论。

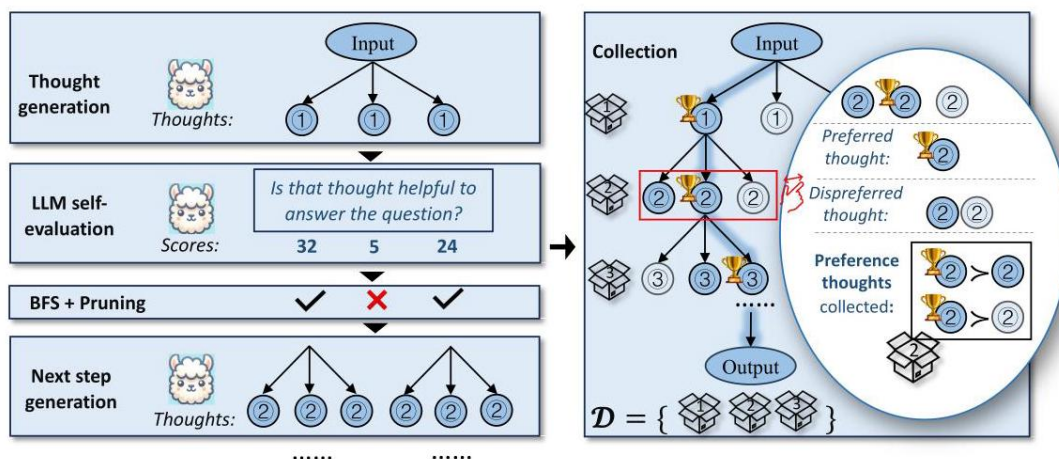


Figure 2: The framework of our CPO method. The left part illustrates the process of generating, evaluating, and pruning thoughts, while the right part demonstrates the collection of preference thoughts. The shaded path represents the final selected reasoning path. Thoughts marked with a trophy indicate preferred data, while sibling nodes without a trophy are marked as dispreferred.

图 2: 我们的 CPO 方法的框架。左半部分说明了思维的生成、评估和修剪过程，而右半部分展示了偏好思维的收集。阴影路径代表最终选择的推理路径。标有奖杯标志的思维表示偏好数据，而没有奖杯标志的兄弟节点则被标记为非偏好。

3.3 Direct Preference Optimization

3.3 直接偏好优化

Direct preference optimization (DPO) is a method for directly optimizing an LLM to align with preference data [13], e.g., human feedback [13, 48, 49]. More specifically, RLHF traditionally frames the application of human feedback to enhance the performance of an LLM within the context of an RL problem. However, DPO reformulates the reward modeling and RL fine-tuning phases in RLHF to a single optimization problem. The objective function of DPO aims to maximize the ratio of probabilities for the preferred responses and optimize the LLM to imitate human preferences.

直接偏好优化 (DPO) 是一种直接优化大语言模型以使其与偏好数据 [13](例如, 人类反馈 [13, 48, 49]) 对齐的方法。更具体地说, 基于人类反馈的强化学习 (RLHF) 传统上在强化学习问题的背景下构建人类反馈的应用, 以提高大语言模型的性能。然而, DPO 将 RLHF 中的奖励建模和强化学习微调阶段重新表述为一个单一的优化问题。DPO 的目标函数旨在最大化偏好响应的概率比, 并优化大语言模型以模仿人类偏好。

Given the generations $(\hat{y}_1, \hat{y}_2) \sim \pi(\hat{y} | x)$ conditioned on input x , these pairs are evaluated and ranked according to specific criteria. Preference data is then constructed from these ranked pairs, denoted by $\hat{y}_w > \hat{y}_l | x$, where $\hat{y}_w$ and $\hat{y}_l$ denote the preferred (winning) and dispreferred (losing) completions between $\hat{y}_1$ and $\hat{y}_2$, respectively. The DPO objective is formulated as follows:

给定以输入 x 为条件的各代 $(\hat{y}_1, \hat{y}_2) \sim \pi(\hat{y} | x)$, 根据特定标准对这些配对进行评估和排序。然后从这些排序后的配对中构建偏好数据, 用 $\hat{y}_w > \hat{y}_l | x$ 表示, 其中 $\hat{y}_w$ 和 $\hat{y}_l$ 分别表示 $\hat{y}_1$ 和 $\hat{y}_2$ 之间的偏好 (获胜) 和非偏好 (失败) 完成情况。DPO 目标的公式如下:

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\log \sigma \left(\beta \log \frac{\pi_{\theta}(\hat{y}_w | x)}{\pi_{\text{ref}}(\hat{y}_w | x)} - \beta \log \frac{\pi_{\theta}(\hat{y}_l | x)}{\pi_{\text{ref}}(\hat{y}_l | x)} \right), \quad (1)$$

where σ is the logistic function, the hyperparameter β regulates the penalty imposed for the deviations from the base reference model π_{ref} .

其中 σ 是逻辑函数, 超参数 β 调节对偏离基础参考模型 π_{ref} 所施加的惩罚。

4 Our Method: Chain of Preference Optimization

4 我们的方法: 偏好优化链

Unlike previous methods that train LLMs to learn the complete reasoning paths [10, 50, 11, 12], our approach leverages the preferences over thoughts generated at each reasoning step, which are often discarded in prior works. Our key insight is that non-optimal thoughts generated during the tree-search process in ToT provide valuable preference information that can enhance LLM's reasoning ability. A major advantage of our method is that it utilizes this supervision only during training, thereby avoiding high inference latency. Our approach consists of two components: synthesizing the chain of preference thoughts (i.e., the preference thoughts in a chain fashion) and training with the CPO objective.

与之前训练大语言模型以学习完整推理路径 [10, 50, 11, 12] 的方法不同，我们的方法利用了在每个推理步骤生成的思维上的偏好，而这些偏好在先前的工作中常常被丢弃。我们的关键见解是，在思维树搜索过程中生成的非最优思维提供了有价值的偏好信息，可以增强大语言模型的推理能力。我们方法的一个主要优点是，它仅在训练期间利用这种监督，从而避免了高推理延迟。我们的方法由两个部分组成：合成偏好思维链（即以链的方式排列的偏好思维）和使用 CPO 目标进行训练。

4.1 Synthesizing the Chain of Preference Thoughts

4.1 合成偏好思维链

Our procedure for synthesizing and collecting preference thought pairs closely follows the inference process of ToT [8]. An overview of our method is shown in Figure 2. Specifically, the detailed process is divided into three parts: 1) thought generation, which generates multiple thoughts for each reasoning step; 2) state evaluation, which evaluates each thought; and 3) search and collection, which finalizes the preference thoughts.

我们合成和收集偏好思维对的过程紧密遵循思维树搜索 (ToT)[8] 的推理过程。我们方法的概述如图 2 所示。具体来说，详细过程分为三个部分：1) 思维生成，为每个推理步骤生成多个思维；2) 状态评估，评估每个思维；3) 搜索和收集，确定偏好思维。

Thought generation. Given a state $s_{i-1} = [x, z_1, \dots, z_{i-1}]$ representing a partial solution with the input x and the sequence of thoughts $[z_1, \dots, z_{i-1}]$ so far, we sample k thoughts for the next reasoning step:

思维生成。给定一个表示部分解决方案的状态 $s_{i-1} = [x, z_1, \dots, z_{i-1}]$ ，其具有输入 x 和到目前为止的思维序列 $[z_1, \dots, z_{i-1}]$ ，我们为下一个推理步骤采样 k 个思维：

$$z_i^{(j)} \sim \pi_\theta(z_i | s_{i-1}) = \pi_\theta(z_i | x, z_1, \dots, z_{i-1}), \text{ for } j = 1, \dots, k. \quad (2)$$

Conditioned on the initial input x , which contains the demonstration examples and the question to be answered, and the previous thoughts $z_1, z_2, \dots, z_{i-1}$, the LLM generates multiple thoughts for the next reasoning step. Specifically, it follows the format of demonstrations, starting with the prefix "Step i ," and samples k thoughts $\{z_i^{(j)}\}_{j=1}^k$. We control the model to pause at the end of $z_i^{(j)}$ by setting the generation of the string "Step $i+1$," as the stop criteria.² As a result, we obtain k new states $s_i^{(j)} = [x, z_1, \dots, z_{i-1}, z_i^{(j)}]$ for $j = 1, \dots, k$.

以包含示范示例和待回答问题的初始输入 x 以及先前的思维 $z_1, z_2, \dots, z_{i-1}$ 为条件，大语言模型为下一个推理步骤生成多个思维。具体来说，它遵循示范的格式，以前缀“步骤 i ”开头，并采样 k 个思维 $\{z_i^{(j)}\}_{j=1}^k$ 。我们通过将字符串“步骤 $i+1$ ”的生设置为停止标准，来控制模型在 $z_i^{(j)}$ 结束时暂停。² 结果，我们为 $j = 1, \dots, k$ 获得 k 个新状态 $s_i^{(j)} = [x, z_1, \dots, z_{i-1}, z_i^{(j)}]$ 。

State evaluation. Given different states $\{s_i^{(j)}\}_{j=1}^k$, we utilize the LLM to reason about the states and evaluate their progress toward solving the problem, eliminating the need for an external reward model or human annotations. To evaluate state $s_i^{(j)}$, the input to the LLM includes specific demonstration examples for the

evaluation process, the input question x , and all the thoughts in the state (i.e., $[z_1, \dots, z_{i-1}, z_i^{(j)}]$). The LLM follows the format of demonstrations to generate a verbal justification first, followed by a classification result from two classes: likely and impossible. The classification results are then used to assign a score, with likely = 10 and impossible = 1.

状态评估。给定不同状态 $\{s_i^{(j)}\}_{j=1}^k$ ，我们利用大语言模型 (LLM) 对这些状态进行推理，并评估它们在解决问题方面的进展，从而无需外部奖励模型或人工标注。为了评估状态 $s_i^{(j)}$ ，LLM 的输入包括评估过程的特定示范示例、输入问题 x 以及该状态下的所有思考内容 (即 $[z_1, \dots, z_{i-1}, z_i^{(j)}]$)。LLM 按照示范示例的格式首先生成文字说明，然后从“可能”和“不可能”两个类别中生成分类结果。然后使用这些分类结果来分配分数，“可能”为 = 10，“不可能”为 = 1。

The prompt template used in our evaluation consists of two parts: (1) the general guidelines, and (2) task-specific demonstration examples. To minimize the effects of randomness and bias, we shuffle the order of demonstration examples [51] and repeatedly sample the generated justification and evaluation results. We then calculate the average score for the state $s_i^{(j)}$. The general guideline prompt for the evaluation is as follows: Evaluate whether the thought helps in partially or directly answering the original question (likely/impossible).

我们评估中使用的提示模板由两部分组成:(1) 一般指南，以及 (2) 特定任务的示范示例。为了尽量减少随机性和偏差的影响，我们打乱示范示例的顺序 [51]，并对生成的说明和评估结果进行重复采样。然后我们计算状态 $s_i^{(j)}$ 的平均分数。评估的一般指南提示如下: 评估该思考是否有助于部分或直接回答原始问题 (可能/不可能)。

Search and collection. We use BFS with pruning as the search algorithm to select the reasoning paths. After evaluation, we retain the n-best thoughts with the highest evaluation scores and proceed to the next step of generation. When the LLM generates a thought containing "so the final answer is:", the search algorithm concludes and returns the selected paths.

搜索与收集。我们使用带剪枝的广度优先搜索 (BFS) 作为搜索算法来选择推理路径。评估后，我们保留评估分数最高的前 n 个最佳思考内容，并进入生成的下一步。当 LLM 生成包含“所以最终答案是:”的思考内容时，搜索算法结束并返回所选路径。

As shown in the right part of Figure 2, after finalizing the reasoning paths, the thoughts within the selected paths are marked as preferred (i.e., winning) thoughts. For each preferred thought at the i -th step z_i^w , we construct corresponding dispreferred (i.e., losing) thoughts. First, we identify the parent state s_{i-1}^w , which includes all the previous thoughts leading to z_i^w . Each child thought of s_{i-1}^w that is not included in the selected path is chosen as a dispreferred thought z_i^l compared to z_i^w . This process results in the preference pair (z_i^w, z_i^l) for the state s_{i-1}^w . We highlight that the constructed dataset $\mathcal{D}$ includes preference data at every step of the reasoning chain. This per-step paired preference supervision is usually overlooked in previous methods [11, 12].

如图 2 右侧所示, 确定推理路径后, 所选路径中的思考内容被标记为首选 (即获胜) 思考内容。对于第 i 步 z_i^w 的每个首选思考内容, 我们构建相应的非首选 (即失败) 思考内容。首先, 我们确定父状态 s_{i-1}^w , 它包括导致 z_i^w 的所有先前思考内容。与 z_i^w 相比, s_{i-1}^w 中未包含在所选路径中的每个子思考内容都被选为非首选思考内容 z_i^l 。这个过程产生了状态 s_{i-1}^w 的偏好对 (z_i^w, z_i^l) 。我们强调, 构建的数据集 $\mathcal{D}$ 包括推理链中每个步骤的偏好数据。这种每一步的配对偏好监督在以前的方法 [11, 12] 中通常被忽视。

4.2 Training with the CPO Objective

4.2 使用 CPO 目标进行训练

Once we have obtained the chain of preference thoughts $\mathcal{D}$, we can proceed with optimization. For the i -th step, given the previous reasoning thoughts s_{i-1}^w , the probabilities of generating z_i^w and z_i^l are denoted as $\pi_\theta(z_i^w | x, s_{i-1}^w)$ and $\pi_\theta(z_i^l | x, s_{i-1}^w)$, respectively. To optimize the LLM on this pair of preference thoughts, we can directly substitute it into Equation 1:

一旦我们获得了偏好思考链 $\mathcal{D}$, 就可以进行优化。对于第 i 步, 给定先前的推理思考内容 s_{i-1}^w , 生成 z_i^w 和 z_i^l 的概率分别表示为 $\pi_\theta(z_i^w | x, s_{i-1}^w)$ 和 $\pi_\theta(z_i^l | x, s_{i-1}^w)$ 。为了在这对偏好思考内容上优化 LLM, 我们可以直接将其代入方程 1:

$$\mathcal{L}_i(\pi_\theta; \pi_{\text{ref}}) = -\log \sigma \left(\beta \log \frac{\pi_\theta(z_i^w | x, s_{i-1}^w)}{\pi_{\text{ref}}(z_i^w | x, s_{i-1}^w)} - \beta \log \frac{\pi_\theta(z_i^l | x, s_{i-1}^w)}{\pi_{\text{ref}}(z_i^l | x, s_{i-1}^w)} \right). \quad (3)$$

² The "stop criteria" is used to control when generation should stop, which is implemented via a function input in Hugging Face's Transformers Library.

² “停止标准”用于控制生成何时停止, 它是通过 Hugging Face 的 Transformers 库中的函数输入实现的。

Thus, the objective function for CPO can be formulated as follows:

因此, CPO 的目标函数可以表述如下:

$$\mathcal{L}_{\text{CPO}}(\pi_\theta; \pi_{\text{ref}}) = \mathbb{E}_{(x, z_i^w, z_i^l, s_{i-1}^w) \sim \mathcal{D}} [\mathcal{L}_i(\pi_\theta; \pi_{\text{ref}})]. \quad (4)$$

5 Experiments

5 实验

In this section, we empirically validate that CPO improves the reasoning ability of the base model, and uncover several insightful findings.

在本节中，我们通过实证验证 CPO 提高了基础模型的推理能力，并发现了一些有见地的结果。

5.1 Settings

5.1 设置

Datasets and evaluation metrics. We focus our research on three types of reasoning tasks: Question Answering (QA), Fact Verification, and Arithmetic Reasoning. For QA, we conduct experiments on three widely used datasets: Bamboogle [17], WikiMultiHopQA [52], and HotpotQA [53]. For fact verification, we use three datasets: Fever [54], Feverous [55], and Vitaminc [56]. For arithmetic reasoning, we test on the SVAMP dataset [57]. Our choice of tasks was driven by the performance of the models using the ToT method, which showed improvements in QA, fact verification, and arithmetic reasoning. We use 4-shot prompting for each dataset, with CoT demonstrations manually constructed by previous works [1, 17, 2]. Detailed experimental configurations can be found in Appendix A. For evaluation metrics, we report the accuracy and the average latency of generating the answer per instance. More metrics can be found in Appendix B.

数据集和评估指标。我们将研究重点集中在三种推理任务上: 问答 (QA)、事实验证和算术推理。对于问答任务，我们在三个广泛使用的数据集上进行实验: Bamboogle [17]、WikiMultiHopQA [52] 和 HotpotQA [53]。对于事实验证，我们使用三个数据集: Fever [54]、Feverous [55] 和 Vitaminc [56]。对于算术推理，我们在 SVAMP 数据集 [57] 上进行测试。我们对任务的选择是由使用思维树 (ToT) 方法的模型性能驱动的，该方法在问答、事实验证和算术推理方面都有改进。我们对每个数据集使用 4 次射击提示，思维链 (CoT) 示范由之前的工作手动构建 [1, 17, 2]。详细的实验配置可在附录 A 中找到。对于评估指标，我们报告每个实例生成答案的准确率和平均延迟。更多指标可在附录 B 中找到。

Baselines. To validate the effectiveness of our proposed CPO, we consider the following baselines: 1) CoT [1], which prompts the LLM to generate a series of reasoning steps before producing the final answer. In our experiments, we use CoT with greedy decoding to assess the model’s reasoning capabilities without any tuning. 2) ToT [8], which requires the LLM to explore multiple reasoning paths via tree search before generating the final answer. We use ToT to select reasoning paths and construct datasets to improve LLM’s reasoning ability in the following TS-SFT baseline and our CPO method. 3) TS-SFT [11], which finds reasoning paths through tree search (i.e., ToT in our implementation) and then uses these paths during the supervised fine-tuning (SFT) process (referred to as SFT in Section 5.3 and 6).

基线。为了验证我们提出的 CPO 的有效性，我们考虑以下基线: 1) 思维链 (CoT) [1]，它促使大语言模型 (LLM) 在产生最终答案之前生成一系列推理步骤。在我们的实验中，我们使用带有贪婪解码的思维链来评估模型的推理能力，无需任何调整。2) 思维树 (ToT) [8]，它要求大语言模型通过树搜索探索多个推理路径，然后再生成最终答案。在以下的 TS - SFT 基线 and 我们的 CPO 方法中，我们使用思维树来选择推理路径并构建数据集，以提高大语言模型的推理能力。3) TS - SFT [11]，它通过树搜索 (即我们实现中的思维树) 找到推理路径，然后在监督微调 (SFT) 过程中使用这些路径 (在第 5.3 节和第 6 节中称为 SFT)。

Implementation Details Our experiments are based on widely used LLMs, specifically LLaMA2- 7B/13B [14] and Mistral-7B [15]. For efficient fine-tuning, we use Low-Rank Adaptation (LoRA) adapters [58]. In all experiments, we set the regularization controller β to 0.1, generate 10 new thoughts for each state, and retain the top 5 thoughts after pruning at each step of reasoning. The temperature is set to 0.9 for SVAMP and 0.4 for the other datasets. The learning rates for DPO and SFT are $5e-6$ and $1e-5$, respectively. We use a batch size (with accumulation) of 32 and optimize the LLM with AdamW [59]. For LoRA, the rank is set to 8, and α is set to 16. All experiments are conducted on NVIDIA A100 GPUs. The latency reported in Table 1 is based on a single NVIDIA A100 40GB. Both training and inference are performed using the Accelerate [60] backend. We train the LLMs for 4 epochs with early stopping based on the performance on a randomly sampled validation set. To mitigate the influence of randomness, all experiments are repeated three times with different random seeds, and the average results are reported.

实现细节我们的实验基于广泛使用的大语言模型，具体是 LLaMA2 - 7B/13B [14] 和 Mistral - 7B [15]。为了进行高效微调，我们使用低秩适应 (LoRA) 适配器 [58]。在所有实验中，我们将正则化控制器 β 设置为 0.1，为每个状态生成 10 个新的思维，并在推理的每个步骤修剪后保留前 5 个思维。对于 SVAMP，温度设置为 0.9，对于其他数据集，温度设置为 0.4。DPO 和 SFT 的学习率分别为 $5e-6$ 和 $1e-5$ 。我们使用批量大小 (带梯度累积) 为 32，并使用 AdamW [59] 优化大语言模型。对于 LoRA，秩设置为 8， α 设置为 16。所有实验都在 NVIDIA A100 GPU 上进行。表 1 中报告的延迟基于单个 NVIDIA A100 40GB。训练和推理都使用 Accelerate [60] 后端。我们基于随机采样验证集上的性能，对大语言模型训练 4 个 epoch 并采用早停法。为了减轻随机性的影响，所有实验使用不同的随机种子重复三次，并报告平均结果。

5.2 Overall Results on Reasoning

5.2 推理的总体结果

Table 1 summarizes the performance across various reasoning tasks. We have the following findings:

表 1 总结了各种推理任务的性能。我们有以下发现:

CPO improves LLM’s reasoning ability. As shown in Table 1, CPO enhances the reasoning ability of the base LLM, achieving an average improvement of 4.3% and a maximum improvement of 9.7% across all tasks and LLMs compared to the CoT approach. This indicates that CPO effectively improves the LLM’s reasoning capabilities. Notably, CPO achieves these improvements without requiring additional human-annotated data, which is particularly beneficial in resource-constrained settings.

CPO 提高了大语言模型的推理能力。如表 1 所示，CPO 增强了基础大语言模型的推理能力，与思维链方法相比，在所有任务和大语言模型上平均提高了 4.3%，最大提高了 9.7%。这表明 CPO 有效地提高了大语言模型的推理能力。值得注意的是，CPO 在不需要额外人工标注数据的情况下实现了这些改进，这在资源受限的环境中特别有益。

CPO has a lower latency than ToT but comparable performance. Although ToT consistently improves performance over CoT, it incurs high latency due to the need to generate and evaluate multiple thoughts at each reasoning step during inference. This process produces numerous tokens, resulting

CPO 的延迟比思维树低，但性能相当。尽管思维树始终比思维链提高性能，但由于在推理过程中每个推理步骤需要生成和评估多个思维，它会导致高延迟。这个过程产生大量令牌，导致

Table 1: Experimental results for ToT, CoT, TS-SFT, and our proposed CPO across complex tasks including question answering, fact verification, and arithmetic reasoning are presented. * means significantly better than the best baseline (TS-SFT) with $p < 0.01$. Bold denotes the best method, and the second best if the top method is ToT. in significant computational and memory overhead [61]. In contrast, CPO shifts this computational burden to the training phase, maintaining the low latency of CoT (i.e., 57.5× faster than ToT on average) during inference while providing comparable or superior performance. This demonstrates that our CPO can deliver enhanced reasoning capabilities without compromising efficiency.

表 1: 展示了思维树 (ToT)、思维链 (CoT)、基于提示选择的微调 (TS-SFT) 以及我们提出的协同推理优化 (CPO) 在包括问答、事实验证和算术推理在内的复杂任务中的实验结果。* 表示在 $p < 0.01$ 条件下显著优于最佳基线 (TS-SFT)。加粗表示最佳方法，如果最佳方法是 ToT，则表示第二好的方法。在显著的计算和内存开销方面 [61]。相比之下，CPO 将这种计算负担转移到训练阶段，在推理过程中保持 CoT 的低延迟 (即平均比 ToT 快 57.5×)，同时提供相当或更优的性能。这表明我们的 CPO 可以在不牺牲效率的情况下提供增强的推理能力。

		ToT [8]		CoT [1]		TS-SFT [11]		CPO (ours)	
		Acc. (%)	Latency (s/ins.)	Acc. (%)	Latency (s/ins.)	Acc. (%)	Latency (s/ins.)	Acc. (%)	Latency (s/ins.)
LLaMA2-7B									
Question Answering	Bam.	33.6	1168.4	29.6	37.2	30.4	36.5	32.0*	38.2
	2Wiki.	28.6	847.6	26.3	35.7	27.6	35.5	29.7*	35.7
	Hot.	23.0	1100.7	21.0	45.5	22.7	44.8	24.0*	41.1
Fact Verification	FVR.	47.3	2087.1	45.8	33.8	47.5	34.0	53.2*	36.8
	FVRS.	47.5	2539.5	44.3	40.6	46.0	40.4	49.0*	41.2
	Vita.	50.7	2639.3	47.3	35.9	51.0	40.1	52.7*	40.1
Arithmetic	SVA.	42.7	1861.1	37.7	33.3	43.1	30.2	46.0*	32.1
Average Performance		39.1	1749.1	36.0	37.4	38.3	37.4	40.9*	37.9
LLaMA2-13B									
Question Answering	Bam.	53.8	1318.3	48.0	50.5	50.8	49.6	52.0*	50.3
	2Wiki.	36.3	1097.1	28.3	67.0	29.0	66.5	30.3*	60.4
	Hot.	32.0	1271.0	29.0	65.2	30.3	65.5	30.3	63.8
Fact Verification	FVR.	48.8	3139.8	48.2	45.4	48.8	44.0	49.2	43.9
	FVRS.	51.3	3433.2	50.0	61.1	48.8	58.0	50.7*	68.2
	Vita.	52.5	2933.6	46.3	48.3	49.7	51.5	54.0*	58.6
Arithmetic	SVA.	45.7	2115.3	40.3	46.2	44.6	46.4	50.0*	48.1
Average Performance		45.8	2186.9	41.4	54.8	43.1	54.5	45.2*	56.2
Mistral-7B									
Question Answering	Bam.	46.4	4399.6	41.6	46.2	41.6	45.0	45.6*	47.0
	2Wiki.	28.4	2356.9	26.7	45.1	31.0	44.2	31.7	46.3
	Hot.	30.0	4698.0	28.0	58.4	28.6	56.2	29.4	56.9
Fact Verification	FVR.	61.4	3291.3	57.9	41.8	60.2	41.7	59.9	40.6
	FVRS.	50.5	5537.8	48.0	51.0	49.7	49.5	54.0*	53.0
	Vita.	52.2	4698.2	47.7	44.8	50.3	45.9	53.7*	46.6
Arithmetic	SVA.	66.0	4623.7	65.3	41.4	59.0	41.3	69.3*	44.9
Average Performance		47.8	4229.4	45.0	47.0	45.8	46.3	49.1*	47.9

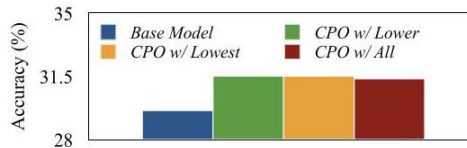
		思维树 [8]		思维链 [1]		TS-SFT [11]		我们的 CPO	
		准确率 (%)	延迟 (秒/英寸)	准确率 (%)	延迟 (秒/英寸)	准确率 (%)	延迟 (秒/英寸)	准确率 (%)	延迟 (秒/英寸)
LLaMA2-7B									
问答	砵。	33.6	1168.4	29.6	37.2	30.4	36.5	32.0*	38.2
	2 维基。	28.6	847.6	26.3	35.7	27.6	35.5	29.7*	35.7
	热。	23.0	1100.7	21.0	45.5	22.7	44.8	24.0*	41.1
事实核查	菲德尔·比利亚尔。	47.3	2087.1	45.8	33.8	47.5	34.0	53.2*	36.8
	FVRS.	47.5	2539.5	44.3	40.6	46.0	40.4	49.0*	41.2
	生命。	50.7	2639.3	47.3	35.9	51.0	40.1	52.7*	40.1
算术	SVA.	42.7	1861.1	37.7	33.3	43.1	30.2	46.0*	32.1
平均性能		39.1	1749.1	36.0	37.4	38.3	37.4	40.9*	37.9
LLaMA2-13B									
问答	砵。	53.8	1318.3	48.0	50.5	50.8	49.6	52.0*	50.3
	2 维基。	36.3	1097.1	28.3	67.0	29.0	66.5	30.3*	60.4
	热。	32.0	1271.0	29.0	65.2	30.3	65.5	30.3	63.8
事实核查	菲德尔·比利亚尔。	48.8	3139.8	48.2	45.4	48.8	44.0	49.2	43.9
	FVRS.	51.3	3433.2	50.0	61.1	48.8	58.0	50.7*	68.2
	生命。	52.5	2933.6	46.3	48.3	49.7	51.5	54.0*	58.6
算术	SVA.	45.7	2115.3	40.3	46.2	44.6	46.4	50.0*	48.1
平均性能		45.8	2186.9	41.4	54.8	43.1	54.5	45.2*	56.2
米斯特拉尔-7B									
问答	砵。	46.4	4399.6	41.6	46.2	41.6	45.0	45.6*	47.0
	2 维基。	28.4	2356.9	26.7	45.1	31.0	44.2	31.7	46.3
	热。	30.0	4698.0	28.0	58.4	28.6	56.2	29.4	56.9
事实核查	菲德尔·比利亚尔。	61.4	3291.3	57.9	41.8	60.2	41.7	59.9	40.6
	FVRS.	50.5	5537.8	48.0	51.0	49.7	49.5	54.0*	53.0
	生命。	52.2	4698.2	47.7	44.8	50.3	45.9	53.7*	46.6
算术	SVA.	66.0	4623.7	65.3	41.4	59.0	41.3	69.3*	44.9
平均性能		47.8	4229.4	45.0	47.0	45.8	46.3	49.1*	47.9

CPO surpasses TS-SFT on average. Despite both CPO and TS-SFT using ToT to generate training data (where our implementation of ToT remains consistent), CPO exhibits an average improvement of 2.7% and reaches a maximum increase of 10.3%. A key factor behind this performance is the CPO’s ability to fully utilize ToT’s reasoning process. Specifically, CPO effectively leverages both selected and unselected thoughts at each reasoning step, whereas TS-SFT only uses information from the selected paths, offering CPO with a clear advantage. A detailed discussion of the effectiveness of CPO is presented in Section 5.3.

平均而言，CPO 超过了 TS-SFT。尽管 CPO 和 TS-SFT 都使用思维树 (ToT) 来生成训练数据 (我们的 ToT 实现保持一致)，但 CPO 平均提高了 2.7%，最大提高了 10.3%。这种性能背后的一个关键因素是 CPO 充分利用 ToT 推理过程的能力。具体来说，CPO 在每个推理步骤中有效地利用了选定和未选定的思维，而 TS-SFT 只使用选定路径中的信息，这使 CPO 具有明显优势。第 5.3 节对 CPO 的有效性进行了详细讨论。

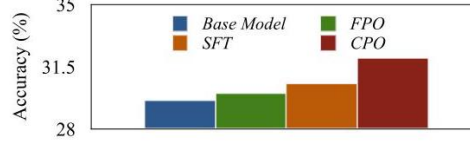
5.3 Component-wise Evaluations

5.3 逐组件评估



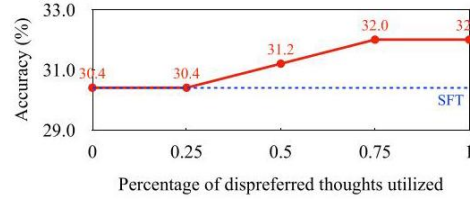
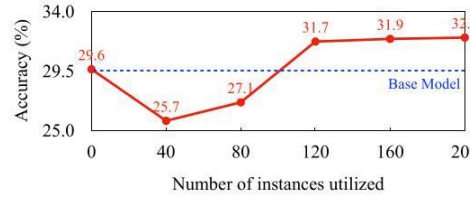
(a) Effect of dispreferred thoughts selection. Base Model: The base LLM. CPO w/ Lowest: CPO using the lowest-scoring thoughts as dispreferred. CPO w/ Lower: CPO using all lower-scoring thoughts as dispreferred. CPO w/ All: CPO using all thoughts not in the selected paths as dispreferred.

(a) 不优选思维选择的影响。基础模型: 基础语言模型。CPO w/ Lowest: 使用得分最低的思维作为不优选思维的 CPO。CPO w/Lower: 使用所有得分较低的思维作为不优选思维的 CPO。CPO w/ All: 使用选定路径之外的所有思维作为不优选思维的 CPO。



(b) Effect of per-step preference supervision. Base Model: The LLM without any fine-tuning. SFT: SFT of the base LLM (i.e., our TS-SFT baseline). FPO: DPO of the base LLM on paired full paths. CPO: CPO of the base LLM on paired per-step thoughts.

(b) 每步偏好监督的影响。基础模型: 未进行任何微调的语言模型。SFT: 基础语言模型的基于思维树的微调 (即, 我们的 TS-SFT 基线)。FPO: 基础语言模型在成对完整路径上的基于偏好的优化。CPO: 基础语言模型在成对每步思维上的基于偏好的优化。



(c) Effect of the number of instances in generating paired thoughts. The number indicates the number of instances (e.g., questions in the QA task) used to generate paired thoughts for our CPO.

(c) 生成成对思维时实例数量的影响。该数字表示用于为我们的 CPO 生成成对思维的实例数量 (例如, 问答任务中的问题)。

(d) Effect of dispreferred thoughts in optimization. The percentage indicates the proportion of data used for CPO, which uses dispreferred thoughts.

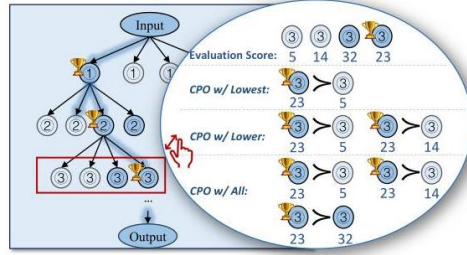
(d) 优化中不优选思维的影响。百分比表示用于使用不优选思维的 CPO 的数据比例。

Figure 3: Component-wise evaluations and analysis on the Bamboogle dataset using the LLaMA2-7B as the base model.

图 3: 以 LLaMA2 - 7B 为基础模型, 在 Bamboogle 数据集上的逐组件评估与分析。

Figure 4: Different strategies for selecting dispre-ferred thoughts and their impact on model performance. At each reasoning step, three strategies are used to select dispreferred thoughts based on their reasoning scores: 1) CPO w/ Lowest: Selects only the thought with the lowest score. 2) CPO w/ Lower: Selects all thoughts with scores lower than the preferred thought. 3) CPO w/All: Selects all thoughts as dispreferred as long as they are not the preferred thought.

图 4: 选择不优选思维的不同策略及其对模型性能的影响。在每个推理步骤中, 根据推理得分使用三种策略选择不优选思维:1)CPO w/ Lowest: 仅选择得分最低的思维。2)CPO w/ Lower: 选择得分低于优选思维的所有思维。3)CPO w/All: 只要不是优选思维, 就将所有思维都选为不优选思维。



Effect of selection methods of dispreferred thoughts. We analyze the impact of different methods for selecting dispreferred thoughts on model performance. As shown in Figure 4, we experiment with three strategies based on evaluation scores for each thought: 1) CPO w/ Lowest: Only the lowest-scoring thoughts in each reasoning step are dispreferred thoughts. 2) CPO w/ Lower: Thoughts with evaluation scores lower than the selected paths are dispreferred thoughts. 3) CPO w/ All: All thoughts not in the selected paths are considered dispreferred thoughts. We ensured an equal number of training samples for each strategy. Note that the evaluation score at each intermediate reasoning step (apart from the final one) determines whether to create the next reasoning step but not which thoughts are preferred. For example, as shown in the figure, even though the score of 32 is higher than 23, the thought with a score of 23 is preferred since it is part of the selected path.

不优选思维选择方法的影响。我们分析了不同的不优选思维选择方法对模型性能的影响。如图 4 所示, 我们基于每个思维的评估得分试验了三种策略:1)CPO w/ Lowest: 在每个推理步骤中, 只有得分最低的思维是不优选思维。2)CPO w/ Lower: 评估得分低于选定路径的思维是不优选思维。3)CPO w/ All: 选定路径之外的所有思维都被视为不优选思维。我们确保每种策略的训练样本数量相等。请注意, 每个中间推理步骤 (除最后一步外) 的评估得分决定是否创建下一个推理步骤, 而不是哪些思维是优选的。例如, 如图所示, 即使 32 分高于 23 分, 但得分 23 分的思维是优选的, 因为它是选定路径的一部分。

The results in Figure 3(a) show that the performance differences among these strategies are minimal. This suggests that the distinction between preferred and dispreferred thoughts is better determined in the selected reasoning path rather than intermediate evaluation scores. To obtain a greater number of preferred thoughts for each instance to create paired preference thoughts, we chose the CPO w/ All strategy.

图 3(a) 中的结果表明, 这些策略之间的性能差异很小。这表明优选和不优选思维之间的区别在选定的推理路径中比在中间评估得分中能更好地确定。为了为每个实例获得更多的优选思维以创建成对偏好思维, 我们选择了 CPO w/ All 策略。

Effect of numbers of training data. To assess the impact of the number of training data used in optimization, we conduct an ablation analysis by varying the number of instances (e.g., questions in the QA task) used to generate paired preference thoughts, ranging from 0 to 200. As illustrated in

训练数据数量的影响。为了评估优化中使用的训练数据数量的影响, 我们通过改变用于生成成对偏好思维的实例数量 (例如, 问答任务中的问题) 进行了消融分析, 范围从 0 到 200。如图所示

Table 2: Effect of different kinds of training data on the Bamboogle dataset using the LLaMA2-7B as the base model.

表 2: 以 LLaMA2 - 7B 为基础模型, 不同类型的训练数据对 Bamboogle 数据集的影响。

Data	Description	SFT	CPO
Single-Task	Training only on specific task (Bamboogle) data.	30.4	32.0
Uniform QA	Training on 3 datasets of the same type (QA) as the test task.	31.2	35.2
Mixed-Type	Training on all 7 different types of data.	29.6	35.2

数据	描述	软件故障排除工具	首席采购官
单任务	仅在特定任务 (Bamboogle) 数据上进行训练。	30.4	32.0
统一问答	在与测试任务相同类型 (问答) 的 3 个数据集上进行训练。	31.2	35.2
混合型	在所有 7 种不同类型的数据上进行训练。	29.6	35.2

Figure 3(c), we observe that with an increase in the number of instances, the model’s performance initially declines and then rises. Specifically, when trained with data generated from less than 80 instances, the model’s performance is even worse than without any training, likely due to overfit-ting [62], which leads to performance degradation. However, as the number increases to 120, the model’s performance consistently improves. Optimizing with paired thoughts from 120 instances, the model’s performance surpasses that of the base model. When the number exceeds 120, the model’s performance converges, indicating a balance of data for training.

在图 3(c) 中, 我们观察到随着实例数量的增加, 模型的性能最初下降然后上升。具体而言, 当使用少于 80 个实例生成的数据进行训练时, 模型的性能甚至比没有任何训练时更差, 这可能是由于过拟合 [62] 导致性能下降。然而, 当数量增加到 120 时, 模型的性能持续提高。使用 120 个实例的配对思维进行优化, 模型的性能超过了基础模型。当数量超过 120 时, 模型的性能收敛, 表明训练数据达到了平衡。

Sensitivity of data mixture. We explore the performance of the CPO method across diverse data settings to assess its adaptability and learning efficiency from various data types. As shown in Table 2, we specifically examine three different data configurations: 1) single task data, 2) uniform QA data, and 3) mixed-type data. Our findings indicate that CPO demonstrates performance improvements of 3.2% in both settings 2 and 3, suggesting its robust ability to harness diverse data sources to enhance learning outcomes. In contrast, the

SFT method exhibits comparable performance across these settings, indicating a different sensitivity to data diversity. It is worth noting that, to ensure fairness, although we find that mixed data leads to better performance, the experiments in Table 1 are conducted using individual datasets for training, consistent with the baselines.

数据混合的敏感性。我们探索 CPO 方法在不同数据设置下的性能，以评估其对各种数据类型的适应性和学习效率。如表 2 所示，我们具体研究了三种不同的数据配置:1) 单任务数据，2) 均匀问答数据，3) 混合类型数据。我们的研究表明，CPO 在设置 2 和设置 3 中均表现出 3.2% 的性能提升，这表明其具有利用多种数据源来提高学习效果的强大能力。相比之下，SFT 方法在这些设置中表现出相当的性能，表明其对数据多样性的敏感性不同。值得注意的是，为确保公平性，尽管我们发现混合数据会带来更好的性能，但表 1 中的实验是使用单个数据集进行训练的，与基线一致。

6 Analysis

6 分析

Do we need dispreferred information? We explore the impact of dispreferred thoughts on model performance by gradually incorporating these thoughts into the training data. Initially, we introduce dispreferred thoughts for their corresponding preferred counterparts and apply CPO to this segment of the data. For preferred thoughts without dispreferred counterparts, we implement SFT on these data. Consequently, the percentage of dispreferred thoughts incorporated can also be viewed as the proportion of data processed using CPO. We adjust the inclusion percentage of dispreferred thoughts from 0% to 100% . An inclusion of 0% indicates that we utilize SFT solely on the preferred thoughts, i.e., the baseline TS-SFT. Conversely, an inclusion of 100% signifies our CPO, where the entire dataset includes paired preferred and dispreferred thoughts.

我们需要非偏好信息吗？我们通过这些思维逐渐纳入训练数据来探索非偏好思维对模型性能的影响。最初，我们为相应的偏好思维引入非偏好思维，并将 CPO 应用于这部分数据。对于没有非偏好对应思维的偏好思维，我们对这些数据实施 SFT。因此，纳入的非偏好思维的百分比也可以视为使用 CPO 处理的数据比例。我们将非偏好思维的纳入百分比从 0% 调整到 100%。纳入 0% 表示我们仅对偏好思维使用 SFT，即基线 TS-SFT。相反，纳入 100% 表示我们使用 CPO，其中整个数据集包括配对的偏好和非偏好思维。

Why is chain level optimization important? As shown in Figure 3(d), we find that increasing the percentage of dispreferred data inclusion consistently improves model performance. This suggests that dispreferred thoughts are beneficial during the optimization process, highlighting the importance of leveraging both preferred and dispreferred thoughts for enhancing the model's reasoning capabilities.

为什么链级优化很重要？如图 3(d) 所示，我们发现增加非偏好数据纳入的百分比会持续提高模型性能。这表明非偏好思维在优化过程中是有益的，突出了利用偏好和非偏好思维来增强模型推理能力的重要性。

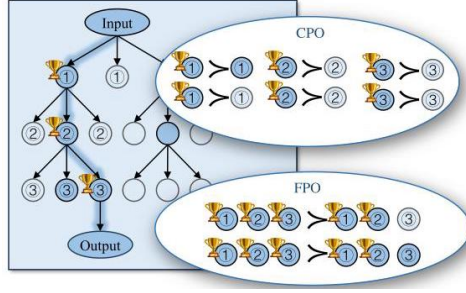


Figure 5: Illustrations of two different ways to construct paired preference data: 1) CPO: Paired preference data are constructed at each thought step. 2) FPO: Paired preference data are constructed only at the full path level.

图 5: 构建配对偏好数据的两种不同方式的图示:(1) CPO: 在每个思维步骤构建配对偏好数据。(2) FPO: 仅在完整路径级别构建配对偏好数据。

Unlike our CPO, an alternative approach is to construct preference data using complete reasoning paths, i.e., using the selected full reasoning paths as preferred and other paths as dispreferred data, as shown in Figure 5. This method essentially applies DPO at the full-path level, referred to here as Full-path Preference Optimization (FPO). However, FPO encounters a significant issue where the gradients of the longest common prefix (LCP) tokens in paired data cancel out, which we call the LCP gradient cancellation issue. For example, for the preferred path $\hat{y}_w = [5, +, 4, =, 9, \text{ and }, 9, +, 2, =, 11]$ and the dispreferred path $\hat{y}_l = [5, +, 4, =, 9, \text{ and }, 9, +, 2, =, 15]$, the gradient will only be computed for the last token where the two sequences diverge.

与我们的 CPO 不同，另一种方法是使用完整推理路径构建偏好数据，即使用选定的完整推理路径作为偏好，其他路径作为非偏好数据，如图 5 所示。这种方法本质上是在完整路径级别应用 DPO，在此称为全路径偏好优化 (FPO)。然而，FPO 遇到了一个重大问题，即配对数据中最长公共前缀 (LCP) 令牌的梯度相互抵消，我们称之为 LCP 梯度抵消问题。例如，对于偏好路径 $\hat{y}_w = [5, +, 4, =, 9, \text{ and }, 9, +, 2, =, 11]$ 和非偏好路径 $\hat{y}_l = [5, +, 4, =, 9, \text{ and }, 9, +, 2, =, 15]$ ，梯度仅会针对两个序列不同的最后一个令牌进行计算。

To mathematically illustrate how LCP gradient cancellation happens in FPO, consider $\hat{y}_w = [p_{1:n}, w_{n+1}]$ and $\hat{y}_l = [p_{1:n}, l_{n+1}]$, where p is the longest common prefix sequence between $\hat{y}_w$ and $\hat{y}_l$. The gradient of FPO is given by:

为了从数学上说明 FPO 中 LCP 梯度抵消是如何发生的，考虑 $\hat{y}_w = [p_{1:n}, w_{n+1}]$ 和 $\hat{y}_l = [p_{1:n}, l_{n+1}]$ ，其中 p 是 $\hat{y}_w$ 和 $\hat{y}_l$ 之间的最长公共前缀序列。FPO 的梯度由下式给出：

$$\nabla_{\theta} \mathcal{L}_{\text{FPO}}(\pi_{\theta}; \pi_{\text{ref}}) = C(\theta) \cdot \nabla_{\theta} (\log \pi_{\theta}(\hat{y}_w | x) - \log \pi_{\theta}(\hat{y}_l | x))$$

$$= C(\theta) \cdot \nabla_{\theta} (\log \pi_{\theta}(p_{1:n} | x) + \log \pi_{\theta}(w_{n+1} | x, p_{1:n}) - \log \pi_{\theta}(p_{1:n} | x) - \log \pi_{\theta}(l_{n+1} | x, p_{1:n})),$$

where $C(\theta)$ is a scalar function that does not affect the direction of the gradient and can be absorbed into the learning rate.

其中 $C(\theta)$ 是一个标量函数，不影响梯度方向，可以被吸收到学习率中。

We can clearly see that the gradient terms of the common prefix tokens (highlighted with boxes) cancel each other out. This issue also exists in DPO training [63], but FPO suffers more frequently and severely due to the longer LCP between paired data constructed by tree search. As an empirical evidence, we observe the LCP length accounts for 28% of the total length in the Bamboogle dataset. CPO, on the other hand, constructs preference data at every step in the reasoning chain, allowing optimization of the LLM on all steps in the reasoning path. This means the common prefix can be optimized at its own step, ensuring that the gradient still exists for the common prefix.

我们可以清楚地看到，公共前缀令牌的梯度项(用方框突出显示)相互抵消。这个问题在 DPO 训练 [63] 中也存在，但由于通过树搜索构建的配对数据之间的最长公共前缀 (LCP) 更长，FPO 更频繁、更严重地受到影响。作为一个经验证据，我们观察到在 Bamboogle 数据集中，LCP 长度占总长度的 28%。另一方面，CPO 在推理链的每一步构建偏好数据，允许在推理路径的所有步骤上对大语言模型进行优化。这意味着公共前缀可以在其自身步骤上进行优化，确保公共前缀的梯度仍然存在。

We also compare FPO to CPO empirically in Figure 3(b), which further substantiates this observation. Switching to FPO led to a relative performance decrease of 4.6%, even worse than the baseline SFT that does not utilize any information from dispreferred data. This underscores the importance of per-step preference thoughts for CPO.

我们还在图 3(b) 中通过实证比较了 FPO 和 CPO，这进一步证实了这一观察结果。切换到 FPO 导致相对性能下降了 4.6%，甚至比不使用任何来自不偏好数据信息的基线 SFT 还要差。这突出了 CPO 中每一步偏好思考的重要性。

7 Conclusion

7 结论

In this work, we introduce a novel method called Chain of Preference Optimization (CPO), which leverages the supervision generated by the self-reasoning process (i.e., tree-of-thoughts) to enhance the reasoning ability of LLMs. Experiments on three different LLMs across seven different datasets demonstrate that CPO can consistently improve the performance of the base model by 4.3% on average without sacrificing inference speed. Furthermore, our method also substantially outperforms the strong baseline TS-SFT and even achieves comparable performance to the ToT method, which requires approximately 57.5 times more inference time.

在这项工作中，我们引入了一种名为偏好优化链 (CPO) 的新方法，该方法利用自我推理过程(即思维树)产生的监督来增强大语言模型的推理能力。在七个不同数据集上对三种不同的大语言模型进行的实验表明，CPO 能够在不牺牲推理速度的情况下，平均将基础模型的性能持续提高 4.3%。此外，我们的方法也显著优于强大的基线 TS-SFT，甚至实现了与 ToT 方法相当的性能，而 ToT 方法的推理时间大约多出 57.5 倍。

In future work, we plan to integrate CPO with other reasoning algorithms, such as Graph-of-Thoughts [9] and AlphaZero-like tree search [11]. Furthermore, we intend to explore the potential of using a weaker

LLM to evaluate a stronger one within the CPO framework, facilitating weak-to-strong alignment [64].

在未来的工作中，我们计划将 CPO 与其他推理算法集成，例如思想图 [9] 和类似 AlphaZero 的树搜索 [11]。此外，我们打算探索在 CPO 框架内使用较弱的语言模型来评估较强语言模型的潜力，以促进从弱到强的对齐 [64]。

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Societal Impacts and Limitations

社会影响与局限性

Since our CPO does not require any human annotation, it can be directly used. For example, to protect the safety of large models, one can simply provide a constitution, and then fine-tune the LLM to make it more compliant. This also introduces another issue: our method can be adjusted for malicious applications. Our limitation is that we still need to generate data through ToT, which is a time-consuming process. We aim to accelerate the complex reasoning processes during training data generation by incorporating methods like non-autoregressive generation [65, 66, 67], speculative decoding [68, 69], and KV cache pruning [70, 71, 72]. Additionally, we have only tested this on text language models and have not tried it on vision-language models. Furthermore, the application scope of our method remains restricted to a small set of downstream tasks. Expanding its application to diverse tasks, such as text classification [73], news verification [74, 75], machine translation [76], and software engineering [77], remains an area for future research. Moreover, ethical considerations must be taken into account, as the potential for misuse could lead to unintended consequences.

由于我们的 CPO 不需要任何人工标注，它可以直接使用。例如，为了保护大模型的安全，只需提供一部宪法，然后对语言模型进行微调，使其更符合安全要求。这也带来了另一个问题：我们的方法可能会被用于恶意应用。我们的局限性在于，我们仍然需要通过思维树 (ToT) 来生成数据，这是一个耗时的过程。我们的目标是通过纳入非自回归生成 [65, 66, 67]、推测性解码 [68, 69] 和键值缓存修剪 [70, 71, 72] 等方法，加速训练数据生成过程中的复杂推理。此外，我们仅在文本语言模型上进行了测试，尚未在视觉语言模型上尝试。此外，我们方法的应用范围仍然局限于一小部分下游任务。将其应用扩展到各种任务，如文本分类 [73]、新闻验证 [74, 75]、机器翻译 [76] 和软件工程 [77]，仍然是未来研究的一个领域。此外，必须考虑伦理因素，因为潜在的滥用可能会导致意想不到的后果。

A Detailed Experiment Configurations

详细的实验配置

To maintain a reasonable budget, especially given the high computational demand of ToT, we limit each dataset to a maximum of 300 test samples through random sampling. For datasets that contain less than 300 test samples, we instead use all available samples. For training, we randomly select less than 300 instances from each dataset to construct the preference data pairs, without using the ground-truth labels. This is because we observe that more number of training data does not lead to performance improvement as shown in Section 5.3. In addition, in our experiments, approximately 200 samples (e.g., questions in the QA task) on average could generate about 6,531 preference pairs, suggesting that our CPO requires only a small number of samples by design. Constructing preference data is a time-intensive process. The choice of 300 samples represents a practical trade-off between efficiency and effectiveness, allowing us to manage resources effectively while still achieving noticeable improvements.

为了保持合理的预算，特别是考虑到思维树 (ToT) 对计算资源的高需求，我们通过随机抽样将每个数据集的测试样本限制在最多 300 个。对于包含少于 300 个测试样本的数据集，我们则使用所有可用样本。对于训练，我们从每个数据集中随机选择少于 300 个实例来构建偏好数据对，而不使用真实标签。这是因为我们观察到，如第 5.3 节所示，更多的训练数据并不会带来性能提升。此外，在我们的实验中，平均大约 200 个样本（例如问答任务中的问题）可以生成大约 6531 个偏好对，这表明我们的 CPO 在设计上只需要少量样本。构建偏好数据是一个耗时的过程。选择 300 个样本是效率和效果之间的实际权衡，使我们能够有效地管理资源，同时仍能实现显著的改进。

B Additional Experiments

B 额外实验

CPO benefits from iterative learning. Inspired by the iterative improvements achieved in previous research [28, 78], in this section, we explore whether CPO can be further improved by iterative learning. Specifically, we try two distinct iterative training strategies: 1. SFT+CPO: in iter=0, Start with a base LLM that has not been fine-tuned at all; in iter=1, SFT the base LLM on the reasoning path selected by ToT (base model); in subsequent iterations (iteration > 1), Continue to fine-tune the model using the CPO method, based on the chain of preference thoughts constructed by the model in the previous iterations. and 2. CPO only: in iter=0,

same as iter=0 in SFT+CPO; in subsequent iterations (iteration >0): Only use the CPO method for training in all iterations, similar to the approach in *SFT + CPO* after the first iteration.

CPO 受益于迭代学习。受先前研究 [28, 78] 中迭代改进的启发, 在本节中, 我们探索 CPO 是否可以通过迭代学习进一步改进。具体来说, 我们尝试两种不同的迭代训练策略: 1. SFT+CPO: 在迭代 =0 时, 从完全未经过微调的基础语言模型开始; 在迭代 =1 时, 在思维树 (ToT) 选择的推理路径上对基础语言模型进行监督微调 (基础模型); 在后续迭代 (迭代 >1) 中, 基于模型在前一次迭代中构建的偏好思维链, 继续使用 CPO 方法对模型进行微调。以及 2. 仅 CPO: 在迭代 =0 时, 与 SFT+CPO 中的迭代 =0 相同; 在后续迭代 (迭代 >0) 中: 在所有迭代中仅使用 CPO 方法进行训练, 类似于 *SFT + CPO* 中第一次迭代后的方法。

As shown in Table 3, We find that if use CoT for inference, as the number of iterations increases, the performance of the model gradually improves. In the CPO only setting, the performance improves by 4% after two iterations. However, an intriguing phenomenon is noted: if we use the ToT method for inference on our fine-tuned models, the performance does not consistently rise and sometimes even declines. For instance, in the SFT+CPO setting, after the first round of SFT, the performance with ToT decreased by 2.7%. We hypothesize this may be related to a decrease in the diversity of the model’s outputs after fine-tuning, which reduces the search space for ToT, making it challenging to find better reasoning paths. When the performance of CoT and ToT becomes similar, further fine-tuning of the LLM leads to convergence in the SFT+CPO setting and even a decline in the CPO only setting.

如表 3 所示, 我们发现如果使用思维链 (CoT) 进行推理, 随着迭代次数的增加, 模型的性能逐渐提高。在仅 CPO 设置中, 经过两次迭代后性能提高了 4%。然而, 注意到一个有趣的现象: 如果我们在微调后的模型上使用思维树 (ToT) 方法进行推理, 性能并非持续上升, 有时甚至会下降。例如, 在 SFT+CPO 设置中, 第一轮监督微调 (SFT) 后, 使用思维树 (ToT) 的性能下降了 2.7%。我们推测这可能与微调后模型输出的多样性降低有关, 这减少了思维树 (ToT) 的搜索空间, 使得难以找到更好的推理路径。当思维链 (CoT) 和思维树 (ToT) 的性能变得相似时, 对语言模型的进一步微调会导致 SFT+CPO 设置中的收敛, 甚至在仅 CPO 设置中导致性能下降。

Comparasion with ReST and self-rewarding baseline. The settings of ReST [33] and self-rewarding [28] are different from ours, as discussed in Section 2. These methods rely on either external reward models (ReST) or labeled data (self-rewarding), which makes them not directly comparable to our approach. To ensure a fair comparison with our CPO method, we prompted the LLM itself in the same way as our CPO to serve as the reward model for ReST and as the labeled data

与 ReST 和自我奖励基线的比较。如第 2 节所述, ReST[33] 和自我奖励 [28] 的设置与我们的不同。这些方法要么依赖外部奖励模型 (ReST), 要么依赖标记数据 (自我奖励), 这使得它们与我们的方法无法直接比较。为了确保与我们的 CPO 方法进行公平比较, 我们以与 CPO 相同的方式提示语言模型本身, 以作为 ReST 的奖励模型和作为标记数据

Inference Method	SFT+CPO				CPO only			
	iter=0 (Base)	iter=1 (SFT)	iter=2 (CPO)	iter=3 (CPO)	iter=0 (Base)	iter=1 (CPO)	iter=2 (CPO)	iter=3 (CPO)
CoT	29.6	30.4	30.4	31.2	29.6	32.0	33.6	31.2
ToT	33.6	30.4	31.2	-	33.6	34.4	33.6	-

推理方法	SFT+CPO				仅 CPO			
	迭代 =0(基础)	迭代 =1 (SFT)	迭代 =2 (CPO)	迭代 =3 (CPO)	迭代 =0 (基础)	迭代 =1 (CPO)	迭代 =2 (CPO)	迭代 =3 (CPO)
思维链	29.6	30.4	30.4	31.2	29.6	32.0	33.6	31.2
树状思维	33.6	30.4	31.2	-	33.6	34.4	33.6	-

Table 3: Effects of iterative learning on the Bamboogle dataset using the LLaMA2-7B as the base model.

表 3: 以 LLaMA2-7B 为基础模型，迭代学习对 Bamboogle 数据集的影响。

annotator for self-rewarding, respectively. As shown in Table 4, the results indicate that, on average, our CPO method surpasses both ReST and self-rewarding under this fair comparison setting.

分别用于自我奖励的注释器。如表 4 所示，结果表明，在这种公平比较设置下，我们的 CPO 方法平均优于 ReST 和自我奖励。

Model	Bam.	2wiki	Hotpotqa	Fever	Feverous	Vitaminc	Svamp	Average
LLaMA2-7b								
Rest	30.4	24.0	22.3	45.5	43.9	51.7	42.3	37.2
SelfR	31.2	25.3	21.0	48.8	44.7	51.3	43.0	37.5
CPO	32.0	29.7	24.0	53.2	49.0	52.7	46.0	40.9
LLaMA2-13b								
Rest	48.0	28.3	28.7	46.8	48.0	50.2	44.3	42.0
SelfR	48.0	29.0	30.0	47.8	48.5	51.0	45.3	42.8
CPO	52.0	30.3	30.3	49.2	50.7	54.0	50.0	45.2
Mistral-7b								
Rest	43.2	26.7	27.4	59.5	48.3	49.7	63.3	45.4
SelfR	44.0	28.0	28.1	58.2	48.0	50.0	65.0	45.9
CPO	45.6	31.7	29.4	59.9	54.0	53.7	69.3	49.1

模型	砰。	2wiki	Hotpotqa	发烧	狂热的	维生素 C	蘑菇	平均
LLaMA2 - 7b								
休息	30.4	24.0	22.3	45.5	43.9	51.7	42.3	37.2
SelfR	31.2	25.3	21.0	48.8	44.7	51.3	43.0	37.5
CPO	32.0	29.7	24.0	53.2	49.0	52.7	46.0	40.9
LLaMA2 - 13b								
休息	48.0	28.3	28.7	46.8	48.0	50.2	44.3	42.0
SelfR	48.0	29.0	30.0	47.8	48.5	51.0	45.3	42.8
CPO	52.0	30.3	30.3	49.2	50.7	54.0	50.0	45.2
米斯特拉尔 - 7b								
休息	43.2	26.7	27.4	59.5	48.3	49.7	63.3	45.4
SelfR	44.0	28.0	28.1	58.2	48.0	50.0	65.0	45.9
CPO	45.6	31.7	29.4	59.9	54.0	53.7	69.3	49.1

Table 4: Performance comparison across different datasets.

表 4: 不同数据集的性能比较。

More metrics on QA dataset. We include F1 scores for the three QA tasks in Table 5. The results show that the F1 performance aligns well with the corresponding accuracy for each task.

问答数据集的更多指标。我们在表 5 中列出了三个问答任务的 F1 分数。结果表明，每个任务的 F1 性能与相应的准确率非常吻合。

Ablation studies across datasets and models. Figure 6, 7, and 8 provide ablations and analysis across various models and datasets. The observed trends remain generally consistent across these different settings.

跨数据集和模型的消融研究。图 6、7 和 8 提供了各种模型和数据集的消融和分析。在这些不同的设置中观察到的趋势总体上保持一致。

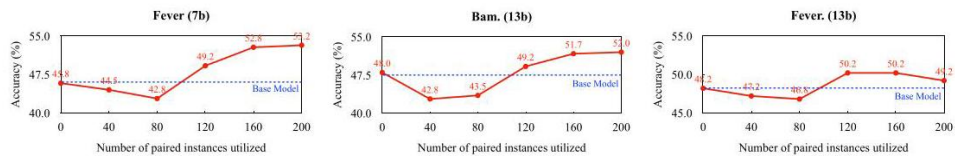


Figure 6: Effect of the number of instances in generating paired thoughts.

图 6: 生成配对思维时实例数量的影响。

Illustrative examples of the reasoning paths preferred by CPO. Table 6 presents examples demonstrating that the paths favored by CPO align more closely with those selected by ToT than by CoT, indicating a higher reasoning quality in CPO's chosen paths.

CPO 偏好的推理路径的示例。表 6 展示了一些示例，表明 CPO 青睐的路径与 ToT 选择的路径比与 CoT 选择的路径更接近，这表明 CPO 选择的路径具有更高的推理质量。

	LLaMA2-7b				LLaMA2-13b				Mistral-7b			
	ToT	CoT	SFT	CPO	ToT	CoT	SFT	CPO	ToT	CoT	SFT	CPO
Bam.	29.5	28.1	29.3	31.8	43.0	40.9	41.4	43.0	48.1	42.2	41.9	47.5
2wiki.	26.5	23.6	25.0	26.0	30.7	26.8	27.2	28.3	25.5	24.3	27.3	27.0
hot.	23.4	20.5	22.9	24.7	33.3	31.3	31.3	32.0	29.9	25.9	26.8	28.0

	LLaMA2-7b				LLaMA2-13b				米斯特拉尔-7b			
	思维树	思维链	监督微调	因果提示优化	思维树	思维链	监督微调	因果提示优化	思维树	思维链	监督微调	因果提示优化
砰。	29.5	28.1	29.3	31.8	43.0	40.9	41.4	43.0	48.1	42.2	41.9	47.5
2 维基。	26.5	23.6	25.0	26.0	30.7	26.8	27.2	28.3	25.5	24.3	27.3	27.0
热。	23.4	20.5	22.9	24.7	33.3	31.3	31.3	32.0	29.9	25.9	26.8	28.0

Table 5: F1 scores on QA datasets.

表 5: 问答数据集上的 F1 分数。

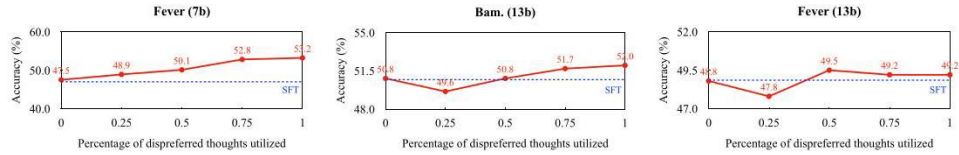


Figure 7: Effect of dispreferred thoughts in optimization.

图 7: 优化中偏好思维的影响。

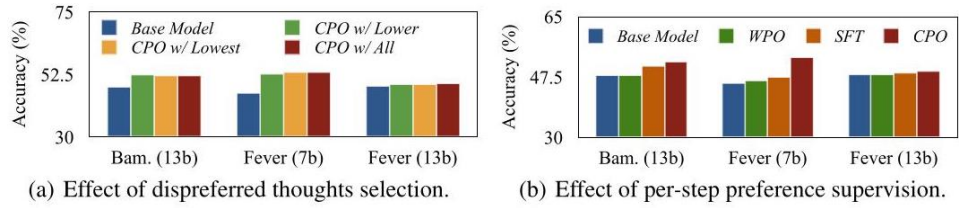


Figure 8: Component-wise Evaluations and Analysis.

图 8: 逐组件评估与分析。

Question 1: When did the author of Annabel Lee enlist in the army?

问题 1: 《安娜贝尔·李》的作者何时参军?

CoT: Step 1, when did Edgar Allan Poe enlist in the army? Edgar Allan Poe enlisted in the army on May 27, 1827. Step 2, when did Edgar Allan Poe die? Edgar Allan Poe died on October 7, 1849. Step 3, so the final answer is: October 7, 1849.

思维链: 步骤 1, 埃德加·爱伦·坡何时参军? 埃德加·爱伦·坡于 1827 年 5 月 27 日参军。步骤 2, 埃德加·爱伦·坡何时去世? 埃德加·爱伦·坡于 1849 年 10 月 7 日去世。步骤 3, 所以最终答案是: 1849 年 10 月 7 日。

ToT: step 1, when did edgar allan poe enlist in the army? edgar allan poe enlisted in the army on august 16, 1827. step 2, what year did edgar allan poe enlist in the army? edgar allan poe enlisted in the army in 1827. step 3, so the final answer is: 1827.

树状思考: 步骤 1, 埃德加·爱伦·坡何时参军? 埃德加·爱伦·坡于 1827 年 8 月 16 日参军。步骤 2, 埃德加·爱伦·坡哪一年参军? 埃德加·爱伦·坡于 1827 年参军。步骤 3, 所以最终答案是: 1827 年。

CPO: step 1, when did Edgar Allan Poe die? Edgar Allan Poe died on October 7, 1849. Step 2, when did Edgar Allan Poe enlist in the army? Edgar Allan Poe enlisted in the army on May 27, 1827. Step 3, so the final answer is: May 27, 1827.

组件优先顺序优化: 步骤 1, 埃德加·爱伦·坡何时去世? 埃德加·爱伦·坡于 1849 年 10 月 7 日去世。步骤 2, 埃德加·爱伦·坡何时参军? 埃德加·爱伦·坡于 1827 年 5 月 27 日参军。步骤 3, 所以最终答案是:1827 年 5 月 27 日。

Question 2: What is the nickname of the easternmost U.S. state?

问题 2: 美国最东部的州的昵称是什么?

CoT: Step 1, who is the scientist at MIT that won the Queen Elizabeth Prize for Engineering in 2013? The scientist at MIT that won the Queen Elizabeth Prize for Engineering in 2013 was Daniela Rus. Step 2, who is the father of Daniela Rus? Daniela Rus's father is Vladimir Rus. Step 3, so the final answer is: Vladimir Rus.

思维链: 步骤 1, 2013 年获得伊丽莎白女王工程奖的麻省理工学院科学家是谁? 2013 年获得伊丽莎白女王工程奖的麻省理工学院科学家是丹妮拉·鲁斯。步骤 2, 丹妮拉·鲁斯的父亲是谁? 丹妮拉·鲁斯的父亲是弗拉基米尔·鲁斯。步骤 3, 所以最终答案是: 弗拉基米尔·鲁斯。

ToT: step 1, what is the name of the easternmost u.s. state? maine is the easternmost u.s. state.step 2, what is the nickname of maine? maine's nickname is 'the pine tree state.

树状思考: 步骤 1, 美国最东部的州叫什么名字? 缅因州是美国最东部的州。步骤 2, 缅因州的昵称是什么? 缅因州的昵称是“松树之州”。

CPO: Step 1, what is the name of the easternmost U.S. state? The easternmost U.S. state is Maine. Step 2, what is the nickname of Maine? The nickname of Maine is The Pine Tree State. Step 3, so the final answer is: The Pine Tree State.

组件优先顺序优化: 步骤 1, 美国最东部的州叫什么名字? 美国最东部的州是缅因州。步骤 2, 缅因州的昵称是什么? 缅因州的昵称是“松树之州”。步骤 3, 所以最终答案是: “松树之州”。

Table 6: Illustrative examples of the reasoning paths preferred by CPO. Similar paths between ToT and CPO are in yellow.

表 6: 组件优先顺序优化偏好的推理路径示例。树状思考和组件优先顺序优化之间的相似路径用黄色表示。

NeurIPS Paper Checklist

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- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.

- 作者应讨论所提出算法的计算效率，以及它们如何随数据集大小进行扩展。

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• 论文中的所有定理、公式和证明都应进行编号并交叉引用。

- All assumptions should be clearly stated or referenced in the statement of any theorems.

• 所有假设都应在任何定理的陈述中明确说明或引用。

- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.

• 证明可以出现在主论文或补充材料中，但如果出现在补充材料中，鼓励作者提供简短的证明概要以提供直观理解。

- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.

• 相反，在论文核心部分提供的任何非正式证明应由附录或补充材料中提供的正式证明进行补充。

- Theorems and Lemmas that the proof relies upon should be properly referenced.

• 证明所依赖的定理和引理应正确引用。

4. Experimental Result Reproducibility

4. 实验结果可重复性

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

问题: 论文是否充分披露了重现论文主要实验结果所需的所有信息，且这些信息会影响论文的主要主张和/或结论 (无论是否提供代码和数据)?

Answer: [Yes]

答案:[是]

Justification: The paper provides a clear and comprehensive explanation of the proposed method. Key hyper-parameters critical for reproducing the results are meticulously detailed.

理由: 该论文对所提出的方法进行了清晰且全面的解释。对重现结果至关重要的关键超参数进行了详细的阐述。

Guidelines:

指南:

- The answer NA means that the paper does not include experiments.

- 答案为“无 (NA)”意味着该论文未包含实验。

- If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.

- 如果论文包含实验, 评审人员可能不会看好对此问题给出“否”的回答: 无论是否提供代码和数据, 使论文具有可重复性都很重要。

- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.

- 如果贡献是一个数据集和/或模型, 作者应描述为使结果具有可重复性或可验证性所采取的步骤。

- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.

- 根据贡献的不同, 可重复性可以通过多种方式实现。例如, 如果贡献是一种新颖的架构, 充分描述该架构可能就足够了; 或者如果贡献是一个特定的模型和实证评估, 可能需要让其他人能够使用相同的数据集复制该模型, 或者提供对该模型的访问权限。一般来说, 发布代码和数据通常是实现这一目标的好方法, 但也可以通过提供如何复制结果的详细说明、对托管模型的访问权限 (例如, 对于大型语言模型的情况)、发布模型检查点或其他适合所进行研究的方式来提供可重复性。

- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example

- 虽然 NeurIPS 不要求发布代码，但会议要求所有提交的论文都要提供某种合理的可重复性途径，这可能取决于贡献的性质。例如

(a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.

(a) 如果贡献主要是一种新算法，论文应明确说明如何重现该算法。

(b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.

(b) 如果贡献主要是一种新的模型架构，论文应清晰且完整地描述该架构。

(c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(c) 如果贡献是一个新模型 (例如，一个大型语言模型)，那么应该有一种方法可以访问该模型以重现结果，或者有一种方法可以重现该模型 (例如，使用开源数据集或关于如何构建数据集的说明)。

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

(d) 我们认识到在某些情况下可重复性可能很棘手，在这种情况下，欢迎作者描述他们提供可重复性的具体方式。对于闭源模型，可能在某种程度上对模型的访问受到限制 (例如，仅限于注册用户)，但其他研究人员应该有可能找到某种途径来重现或验证结果。

5. Open access to data and code

5. 数据和代码的开放获取

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

问题: 如补充材料中所述，论文是否提供对数据和代码的开放获取，并带有足够的说明以忠实地重现主要实验结果？

Answer: [Yes]

答案:[是]

Justification: The code is provided in the supplemental material.

理由: 代码在补充材料中提供。

Guidelines:

指南:

- The answer NA means that paper does not include experiments requiring code.

- 答案“NA”表示该论文不包含需要代码的实验。

- Please see the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.

- 有关更多详细信息, 请参阅 NeurIPS 代码和数据提交指南 (<https://nips.cc/public/guides/CodeSubmissionPolicy>)。

- While we encourage the release of code and data, we understand that this might not be possible, so “No” is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).

- 虽然我们鼓励发布代码和数据, 但我们理解这可能无法实现, 因此“否”是一个可接受的答案。论文不能仅仅因为未包含代码而被拒稿, 除非这对论文的贡献至关重要 (例如, 对于一个新的开源基准测试)。

- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.

- 说明应包含运行以重现结果所需的精确命令和环境。有关更多详细信息, 请参阅 NeurIPS 代码和数据提交指南 (<https://nips.cc/public/guides/CodeSubmissionPolicy>)。

- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.

- 作者应提供数据访问和准备方面的说明, 包括如何访问原始数据、预处理数据、中间数据和生成的数据等。

- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.

- 作者应提供脚本以重现新提出的方法和基线的所有实验结果。如果只有一部分实验可重现, 他们应说明脚本中省略了哪些实验以及原因。

- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- 在提交时, 为了保持匿名性, 作者应发布匿名版本 (如适用)。

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

- 建议在补充材料 (附于论文之后) 中提供尽可能多的信息, 但允许包含数据和代码的网址。

6. Experimental Setting/Details

6. 实验设置/细节

Question: Does the paper specify all the training and test details (e.g., data splits, hyper-parameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

问题: 论文是否详细说明了理解结果所需的所有训练和测试细节 (例如, 数据划分、超参数、它们是如何选择的、优化器类型等)?

Answer: [Yes]

答案: [是]

Justification: The experiment setting and details is presented in Section 5.1 and A.

理由: 实验设置和细节在第 5.1 节和 A 中给出。

Guidelines:

指南:

- The answer NA means that the paper does not include experiments.

- 答案 NA 表示该论文未包含实验。

- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.

- 实验设置应在论文核心部分呈现, 详细程度需达到能够理解结果并使其有意义的必要水平。

- The full details can be provided either with the code, in appendix, or as supplemental material.

- 完整细节可以随代码一同在附录中提供, 或作为补充材料提供。

7. Experiment Statistical Significance

7. 实验的统计显著性

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

问题: 论文是否适当地报告了误差线以及正确定义的或关于实验统计显著性的其他适当信息?

Answer: [Yes]

答案:[是]

Justification: The results are accompanied by significant tests.

理由: 结果附有显著性检验。

Guidelines:

指导原则:

- The answer NA means that the paper does not include experiments.
 - 答案为“不适用 (NA)”意味着论文不包括实验。
- The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
 - 如果结果附有误差线、置信区间或统计显著性检验, 至少对于支持论文主要主张的实验, 作者应回答“是”。
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
 - 应明确说明误差线所捕捉的变异性因素 (例如, 训练/测试分割、初始化、某些参数的随机抽取或在给定实验条件下的整体运行)。
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
 - 应解释计算误差线的方法 (闭式公式、调用库函数、自助法等)。
- The assumptions made should be given (e.g., Normally distributed errors).

- 应给出所做的假设 (例如, 误差呈正态分布)。

- It should be clear whether the error bar is the standard deviation or the standard error of the mean.

- 应明确误差线是标准差还是均值的标准误差。

- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.

- 报告 1 个标准差的误差线是可以的, 但应说明。如果未验证误差正态性假设, 作者最好报告 2 个标准差的误差线, 而不是说明他们有 96% 的置信区间。

- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).

- 对于非对称分布, 作者应注意不要在表格或图表中显示会产生超出范围结果 (例如, 负误差率) 的对称误差线。

- If error bars are reported in tables or plots, The authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

- 如果在表格或图表中报告了误差线, 作者应在文本中解释其计算方法, 并在文本中引用相应的图表。

8. Experiments Compute Resources

8. 实验计算资源

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

问题: 对于每个实验, 论文是否提供了关于重现实验所需的计算机资源 (计算工作者类型、内存、执行时间) 的足够信息?

Answer: [Yes]

答案:[是]

Justification: The information on the computer resources can be found in Section 5.1 and A. Guidelines:

理由: 关于计算机资源的信息可在第 5.1 节和 A. 指南中找到:

- The answer NA means that the paper does not include experiments.

- 答案 NA 表示论文不包括实验。

- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.

- 论文应指明计算工作者的类型 (CPU 或 GPU、内部集群或云提供商), 包括相关内存和存储。

- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.

- 论文应提供每个单独实验运行所需的计算量, 并估计总计算量。

- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

- 论文应披露整个研究项目所需的计算量是否超过论文中报告的实验所需的计算量 (例如, 未纳入论文的初步或失败实验)。

9. Code Of Ethics

9. 道德准则

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

问题: 论文中进行的研究在各方面是否符合 NeurIPS 道德准则
<https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

答案:[是]

Justification: The research conform the NeurIPS Code of Ethics.

理由: 该研究符合 NeurIPS 道德准则。

Guidelines:

指南:

- The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.

- 答案 NA 表示作者未审查 NeurIPS 道德准则。

- If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.

- 如果作者回答否，他们应解释需要偏离道德准则的特殊情况。

- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

- 作者应确保保持匿名 (例如，如果因其所在司法管辖区的法律或法规有特殊考虑)。

10. Broader Impacts

10. 更广泛的影响

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

问题: 论文是否讨论了所开展工作的潜在积极社会影响和消极社会影响?

Answer: [Yes]

答案:[是]

Justification: The paper adequately discusses the impacts of the work in Section 7.

理由: 论文在第 7 节充分讨论了该工作的影响。

Guidelines:

指南:

- The answer NA means that there is no societal impact of the work performed.

- 答案“不适用 (NA)”意味着所开展的工作没有社会影响。

- If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.

- 如果作者回答“不适用 (NA)”或“否”，他们应解释其工作为何没有社会影响或论文为何未涉及社会影响。

- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.

- 消极社会影响的示例包括潜在的恶意或意外用途(例如, 虚假信息、生成虚假个人资料、监视)、公平性考量(例如, 部署可能做出不公平影响特定群体决策的技术)、隐私考量和安全考量。

- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.

- 会议预计许多论文将是基础研究, 与特定应用无关, 更不用说部署了。然而, 如果存在通向任何负面应用的直接途径, 作者应指出。例如, 指出生成模型质量的提高可用于生成虚假信息的深度伪造是合理的。另一方面, 无需指出用于优化神经网络的通用算法可使人们更快地训练生成深度伪造的模型。

- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.

- 作者应考虑技术按预期使用且运行正常时可能出现的危害、技术按预期使用但给出错误结果时可能出现的危害以及(有意或无意)滥用技术导致的危害。

- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

- 如果存在负面社会影响, 作者还可讨论可能的缓解策略(例如, 模型的受限发布、除攻击外提供防御、监测滥用的机制、监测系统如何随时间从反馈中学习的机制、提高机器学习的效率和可及性)。

11. Safeguards

11. 保障措施

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

问题: 论文是否描述了针对有高滥用风险的数据或模型 (例如预训练语言模型、图像生成器或抓取的数据集) 的负责任发布所采取的保障措施?

Answer: [NA]

答案:[不适用 (NA)]

Justification: The paper poses no such risks.

理由: 论文不存在此类风险。

Guidelines:

指南:

- The answer NA means that the paper poses no such risks.
- 答案“不适用 (NA)”意味着论文不存在此类风险。
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- 对于存在被滥用或两用高风险的已发布模型，应在发布时采取必要的安全措施，以实现模型的可控使用，例如要求用户遵守使用指南或访问限制，或实施安全过滤。
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- 从互联网上抓取的数据集可能存在安全风险。作者应描述他们如何避免发布不安全的图像。
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.
- 我们认识到提供有效的安全措施具有挑战性，而且许多论文并不要求这样做，但我们鼓励作者考虑这一点并做出真诚的努力。

12. Licenses for existing assets

12. 现有资产的许可

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

问题: 论文中使用的资产 (如代码、数据、模型) 的创建者或原始所有者是否得到了恰当的认可，许可和使用条款是否被明确提及并得到了恰当的尊重？

Answer: [Yes]

答案:[是]

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Guidelines:

指南:

- The answer NA means that the paper does not use existing assets.

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- The authors should cite the original paper that produced the code package or dataset.

- 作者应引用生成代码包或数据集的原始论文。

- The authors should state which version of the asset is used and, if possible, include a URL.

- 作者应说明使用的是资产的哪个版本, 并尽可能提供网址。

- The name of the license (e.g., CC-BY 4.0) should be included for each asset.

- 应包括每个资产的许可名称 (如 CC - BY 4.0)。

- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.

- 对于从特定来源 (如网站) 抓取的数据, 应提供该来源的版权和服务条款。

- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.

- 如果资产已发布, 应提供包中的许可、版权信息和使用条款。对于流行的数据集, paperswithcode.com/datasets 为一些数据集整理了许可。他们的许可指南可帮助确定数据集的许可。

- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.

- 对于重新打包的现有数据集, 应同时提供原始许可和派生资产的许可 (如果有变化)。

- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

- 如果此信息无法在线获取, 鼓励作者联系资产的创建者。

13.New Assets

13. 新资产

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

问题: 论文中引入的新资产是否有充分记录, 且相关文档是否与资产一同提供?

Answer: [NA]

答案:[无可用信息]

Justification: The paper does not release new assets.

理由: 论文未发布新资产。

Guidelines:

指南:

- The answer NA means that the paper does not release new assets.
- 答案“无可用信息”表示论文未发布新资产。
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- 研究人员应通过结构化模板在提交内容中传达数据集/代码/模型的详细信息。这包括有关训练、许可、限制等的详细信息。
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- 论文应讨论是否以及如何获得了使用其资产的人的同意。
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.
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14. Crowdsourcing and Research with Human Subjects

14. 众包与人体研究

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

问题: 对于众包实验和人体研究, 论文是否包含提供给参与者的完整指导说明文本以及适用情况下的屏幕截图, 以及有关补偿 (如有) 的详细信息?

Answer: [NA]

答案:[无可用信息]

Justification: The paper does not involve crowdsourcing nor research with human subjects. Guidelines:

理由: 论文不涉及众包也不涉及人体研究。指南:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.

- 答案“无可用信息”表示论文不涉及众包也不涉及人体研究。

- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.

- 将此信息包含在补充材料中是可以的, 但如果论文的主要贡献涉及人体研究, 那么应在主论文中包含尽可能多的详细信息。

- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

- 根据 NeurIPS 道德准则, 参与数据收集、整理或其他工作的人员应至少获得数据收集所在国家的最低工资。

15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects

15. 机构审查委员会 (IRB) 对涉及人类受试者研究的批准或同等程序

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

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Answer: [NA]

答案:[不适用]

Justification: The paper does not involve crowdsourcing nor research with human subjects. Guidelines:

理由: 论文不涉及众包, 也不涉及对人类受试者的研究。指导原则如下:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.

- 答案“不适用”表示论文不涉及众包, 也不涉及对人类受试者的研究。

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- 根据研究开展所在的国家/地区, 任何涉及人类受试者的研究可能都需要 IRB 批准 (或同等程序)。如果您获得了 IRB 批准, 应在论文中明确说明。

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